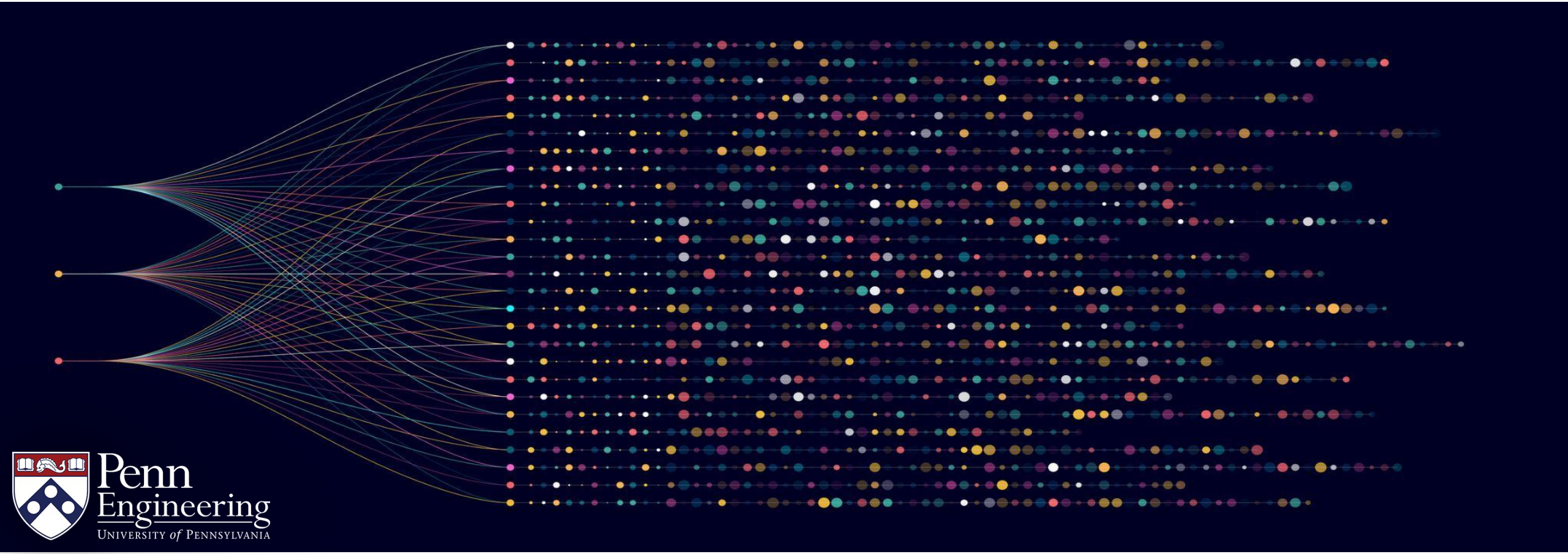
Algorithms for Big Data

Anindya De Erik Waingarten

Estimating Averages: Problem Setup



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Objectives: Computing Large Averages

- Motivating Scenario
- Defining the computational problem
- A “brute force” algorithmic benchmark

Motivating Scenario

- You are a software engineer working for a social network
- There is a (very large) database which holds all of user information and maintains usage information
- You are tasked with determining certain aggregate statistics on users and their usage information

Motivating Scenario - Example Questions

- How many users are in Europe?
- How many users posted in the past week?
- How long (on average) did users spend on the platform today?

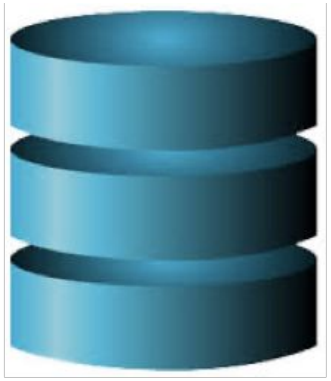
Commonalities: Ask for large averages over the entire dataset

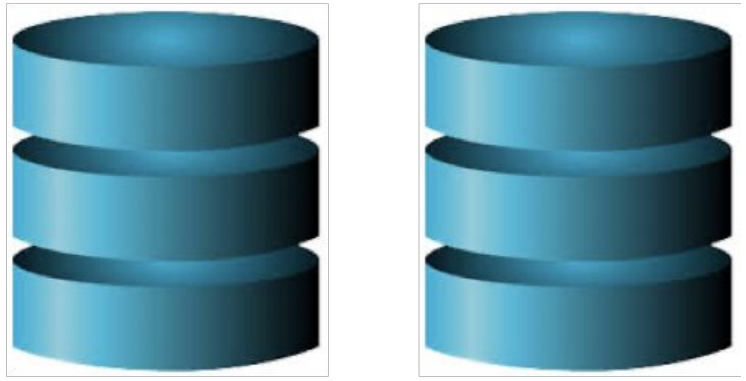
Outline for what is coming:

- First, provide background on the scenario
  - How data is stored
  - How data is accessed
  - What we want to do

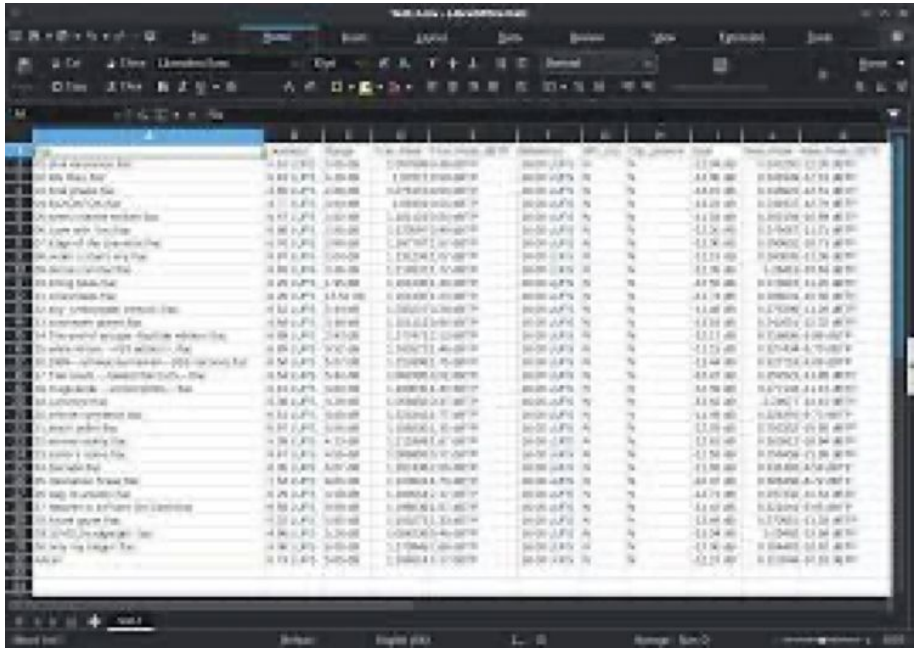
- Then, define computational problem
  - Incorporating elements from the scenario
  - Establish algorithmic benchmarks

Background: What is the data?





Input data
(possibly in the cloud)



Rows may be users/events,
columns are attributes

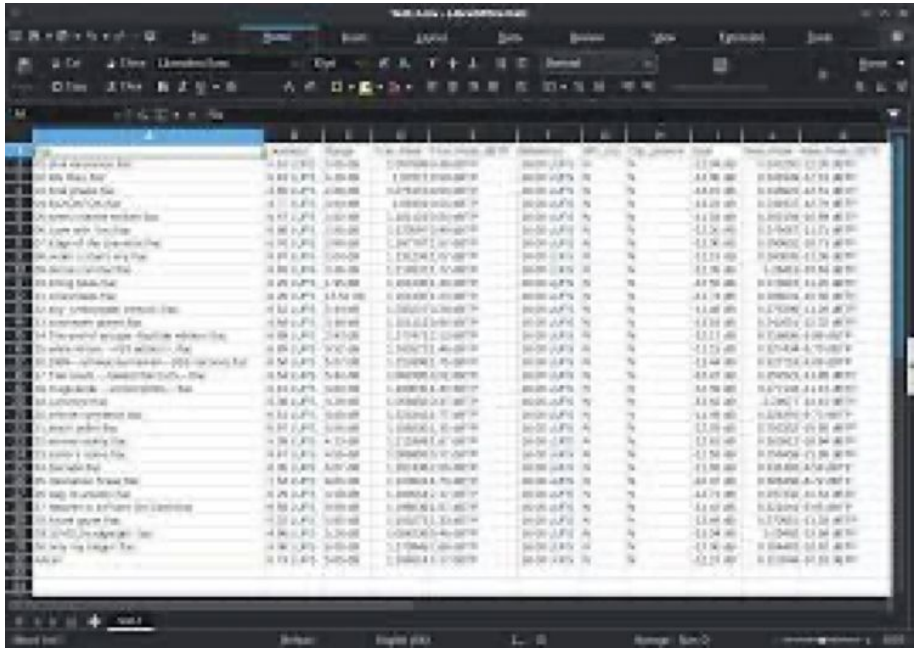
Background: What is the data?

Establish notation:

$$n = \text{ number of items.}$$

$$X_1, \ldots, X_n \in \mathbb{R}$$

$$S = \frac{1}{n} \sum_{i=1}^{n} X_i$$

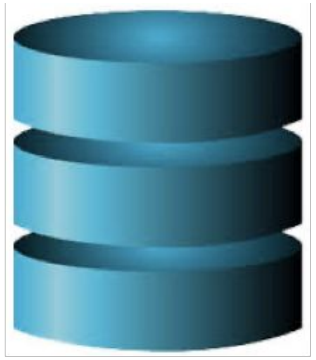


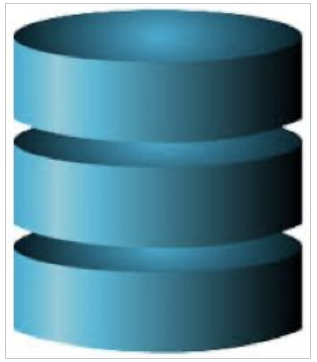
Rows may be users/events, columns are attributes

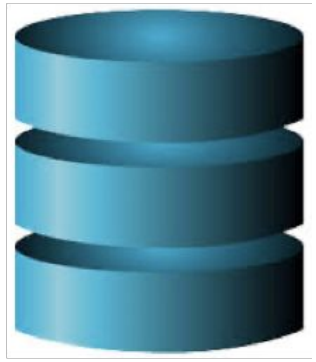
Background: What is the access?

Query

Result of Query







Background: What is the access?

- The input data may not be readily accessible.
  - If data lies in a data center, communication costs may be bottleneck

- Input data is accessible. We want our algorithms to be very fast.
  - The running time of the algorithm is dominated by the number of queries made

Defining the Computational Problem.

Input: $X_1, \ldots, X_n \in \mathbb{R}$

Goal: $S = \frac{1}{n} \sum_{i=1}^{n} X_i$

Queries: X_i

Examples of estimating large averages:

1. How many users are in Europe?
2. How many users posted in the past week?
3. How long did users spend on the platform on average?

Algorithmic Benchmark

Approach: brute force

Algorithm:

- Instantiate a variable which will maintain the sum
- Query every value and maintain sum
- Output the sum divided by n

Questions:

- Is it correct?
- How long should we expect it to take?

Can one be faster?

- Brute force: exact and deterministic
  - downside: slow

- Do there exist faster algorithms?
  - No, if we insist on exact or deterministic algorithms

- If we allow *approximation* and *randomization*, we can design significantly faster algorithms

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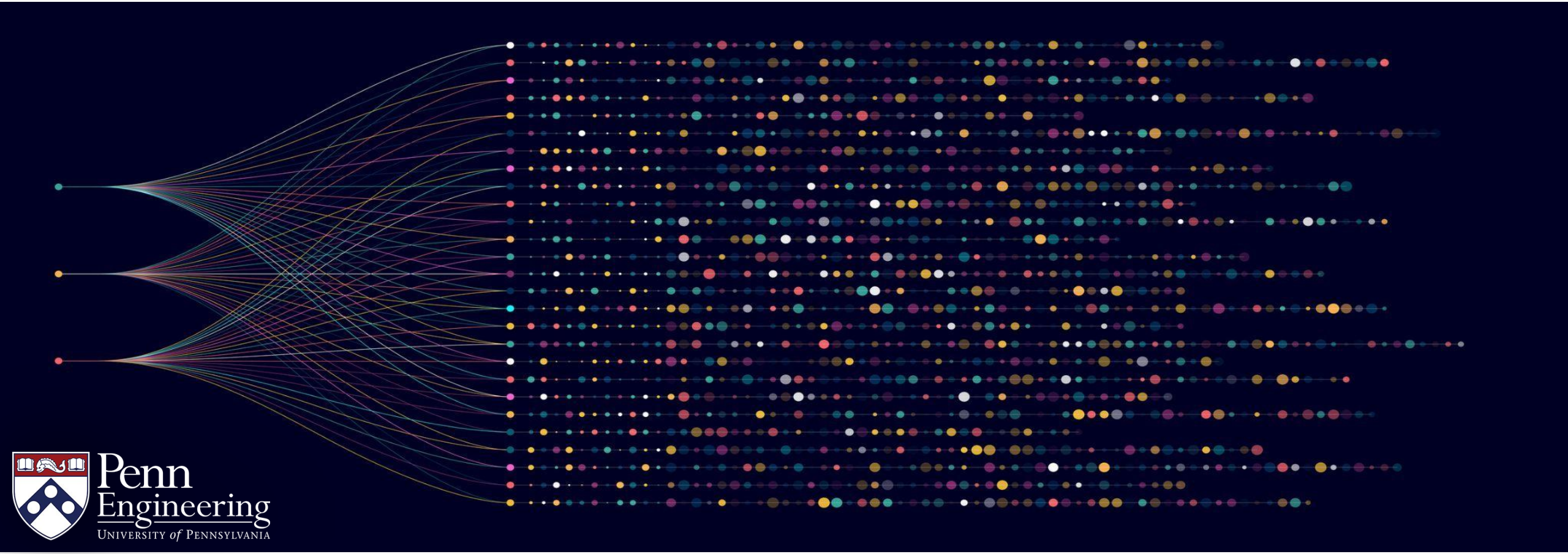
Additive Approximation Guarantees



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Objectives: Computing Large Averages

- Estimating large averages

- Additive approximations

Defining the Computational Problem

Input: $X_1, \ldots, X_n \in \mathbb{R}$

Goal: $S = \frac{1}{n} \sum_{i=1}^{n} X_i$

Queries: X_i

Examples of large averages:

1. How many users are in Europe?
2. How many users posted in the past week?
3. How long did users spend on the platform on average?

When is approximation adequate?

- Would an approximation to the output be sufficient?

- How would the result of the algorithm impact downstream decisions (or other algorithms)?

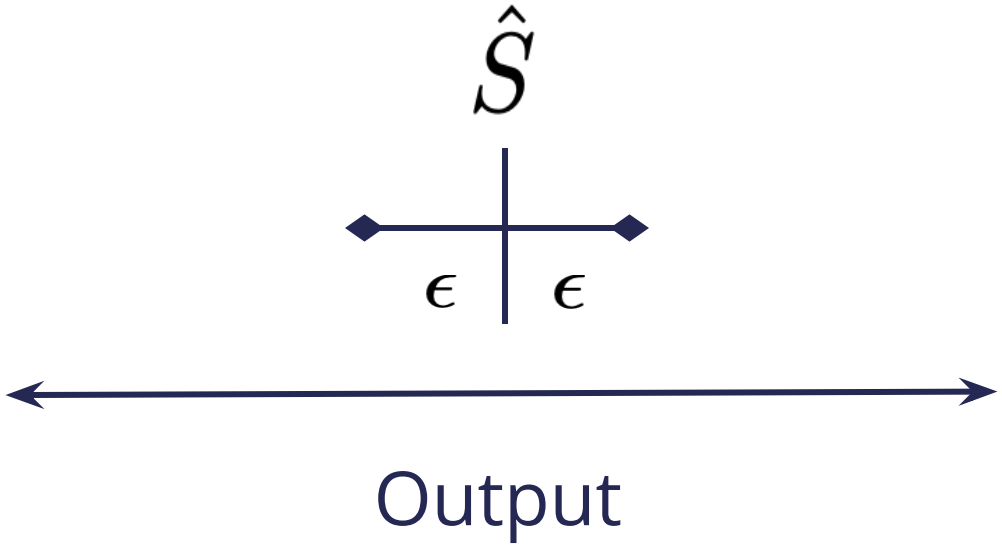
- If input is too large, no choice but to approximate. It is important to know what you are getting.

What do approximation guarantees look like?

- A number approximates another number when they are close

- Execute an algorithm which outputs $\hat{S} \in \mathbb{R}$ such that for a known $\epsilon \geq 0$:

$$|\hat{S} - S| \leq \epsilon$$



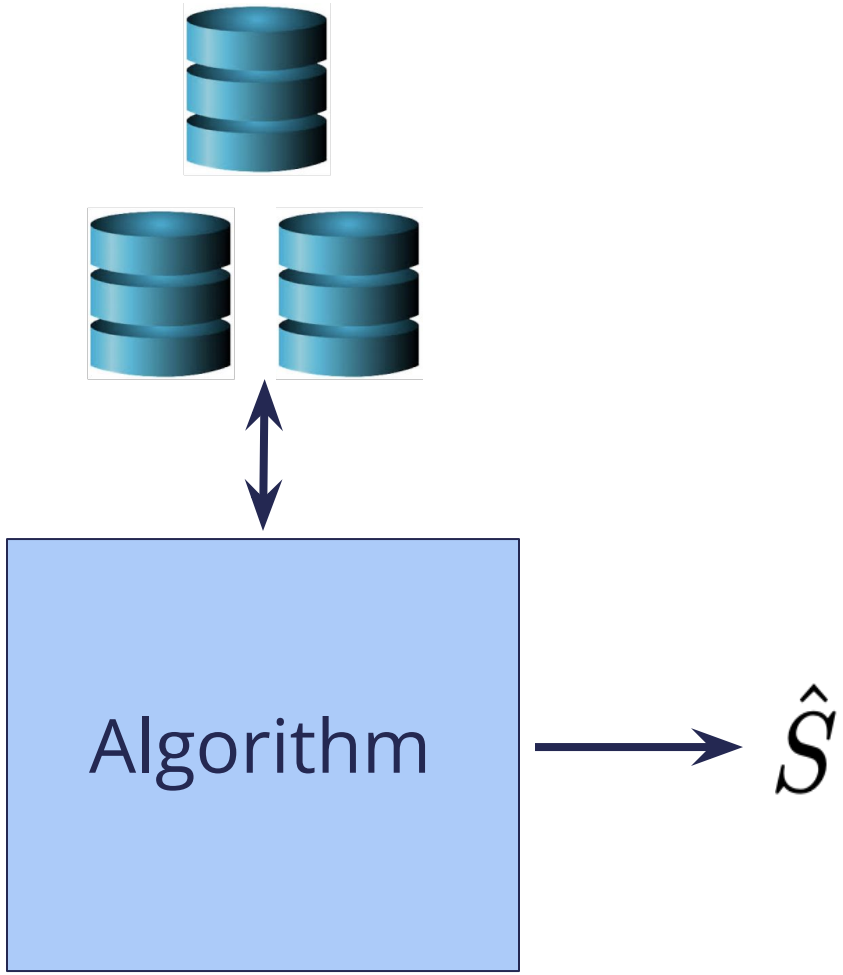
Additive ϵ-approximations

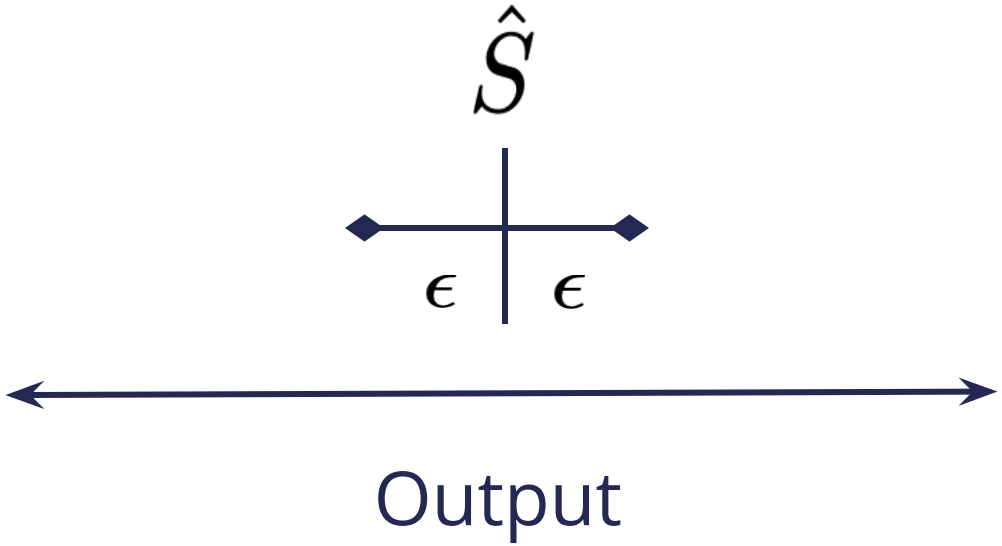
Definition: An algorithm for computing a large average is an additive ϵ-approximation if it satisfies the following guarantees:

- Given input query access to $X_1, \ldots, X_n \in \mathbb{R}$
- It produces an output number $\hat{S}$ which satisfies

$$\left| \hat{S} - \frac{1}{n} \sum_{i=1}^{n} X_i \right| \leq \epsilon.$$

What do approximation guarantees look like?





Interpreting results of approximations

- How many users are in Europe?
  - If accuracy is 0.05, and output is 0.3, then between 25% and 35% of users are in Europe
- How many users posted in the past week?
  - If accuracy is 0.1 and output is 0.9, then between 80% and 100% of users posted in the past week
- How long (on average) did users spend on the platform today?
  - If accuracy is 30 and output is 120, then between 90 and 150 seconds

Approximation guarantees

- Accuracy parameter $\epsilon \geq 0$. As ϵ becomes smaller, the algorithm becomes more accurate. When zero, algorithm is exact.

- As the algorithm becomes more accurate, the number of queries grows. More accurate algorithms are more expensive.

- Big sacrifice. Algorithm will no longer be exactly correct.

Key algorithmic questions:

- Do there exist (more) efficient algorithms when allowing approximation?
  - By more efficient, fewer than $O(n)$ queries, which was the brute force benchmark.

- If so, what is the tradeoff between the number of queries and the accuracy parameter.

- Turns out no, unless algorithm is also randomized.

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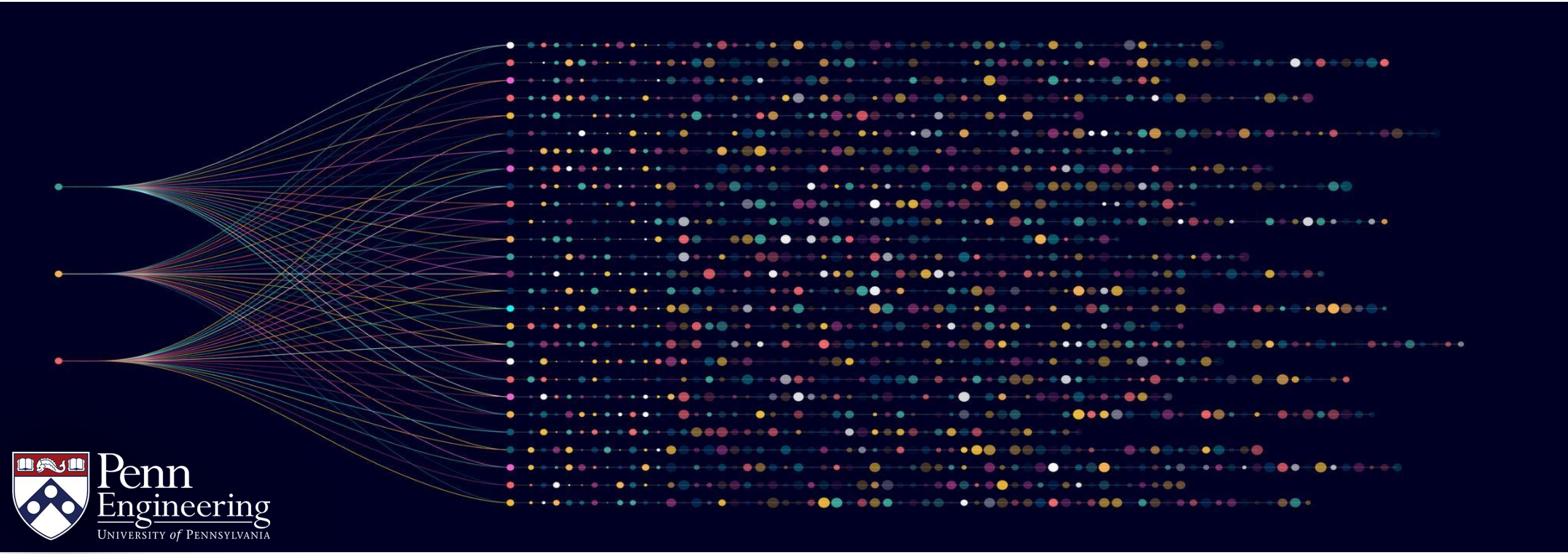
Pitfall of Deterministic Algorithms



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Objectives: Computing Large Averages

- Randomized algorithms for large averages

- Why deterministic algorithms cannot produce accurate approximations

Defining the Computational Problem.

Input: $X_1, \ldots, X_n \in \mathbb{R}$ and $\epsilon \geq 0$

Goal: Output $\hat{S}$ which satisfies $\left| \hat{S} - \frac{1}{n} \sum_{i=1}^{n} X_i \right| \leq \epsilon.$

Bad news: Deterministic algorithm must make linearly many queries

What is randomization?

- Randomized algorithm output random variables
  - Algorithm “flips coins” internally, specific output is inherently unpredictable

- For approximate guarantees with sublinearly many queries, randomness is unavoidable

Why is randomness necessary?

Input variables:

0	0	1	1	1	1	0	1	0	1	1	0	0	0	1	1

A deterministic algorithm (which does not use any randomness) is a rule for selecting queries

Goal: two inputs with different averages which look the same to a deterministic algorithm

We construct two examples to “fool” algorithm

Input variables:

Assume a deterministic algorithm making 5 queries.

- When query position i, make that 0

Possible Inputs:

	Average
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0
0 0 1 1 0 1 1 1 1 0 1 1 1 1 1 0	11/16

Precise bounds for the “fooling set” argument

- Fix a value of n and a deterministic algorithm making q queries
- Two inputs:
  - All 0’s input — Average = 0
  - The q queries are 0, rest are 1’s — Average = 1 - q/n
- Best case scenario: $\frac{1}{2}(1 - q/n) \leq \epsilon$.
- Conclusion: queries must be at least $n(1 - 2\epsilon)$

The pitfall of deterministic algorithms

- Deterministic algorithms commit to a fixed action
- If the number of queries is small, there are many values which were not observed
- There are two different inputs which are indistinguishable to the algorithm, but have significantly different answers

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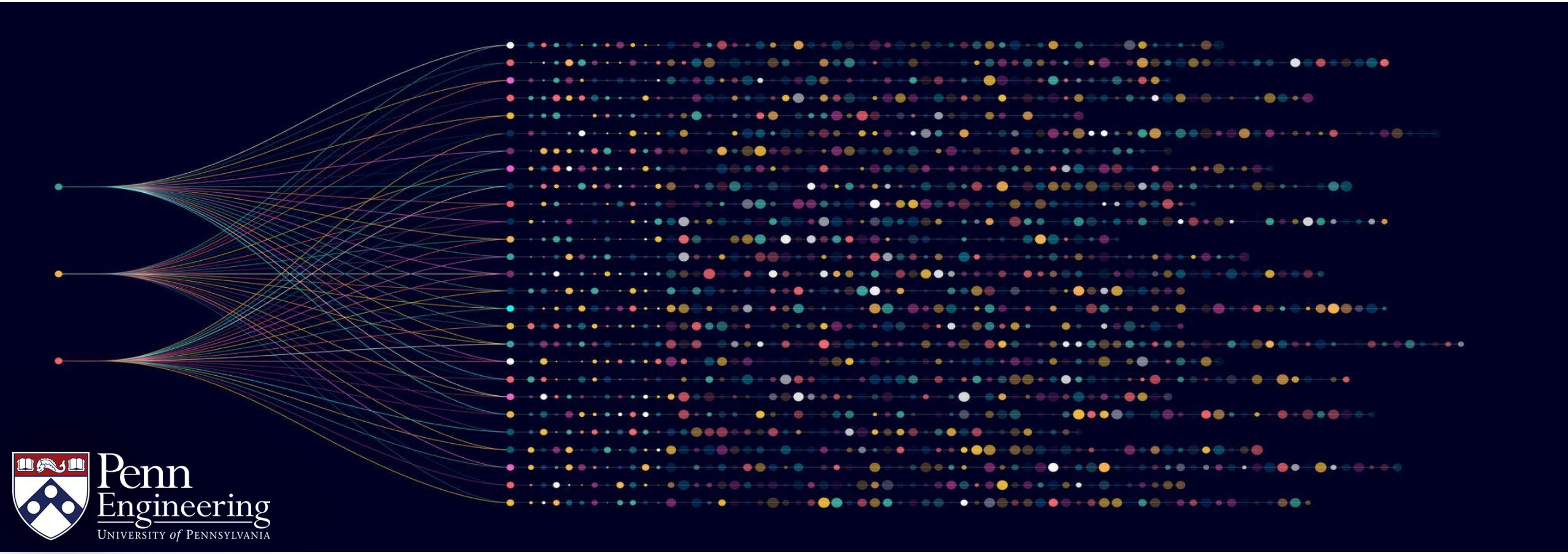
Randomization: Pros and Cons



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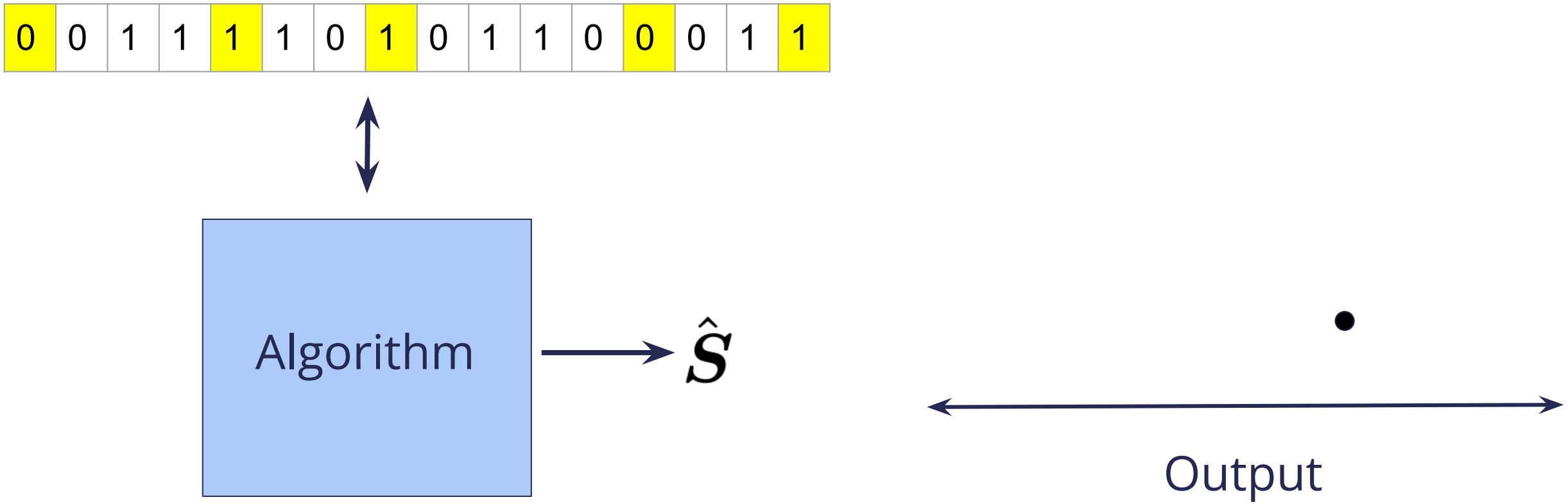
Objectives: Computing Large Averages

- Randomized algorithms

- Pros and cons of randomization

What if the algorithm does not commit to certain queries?



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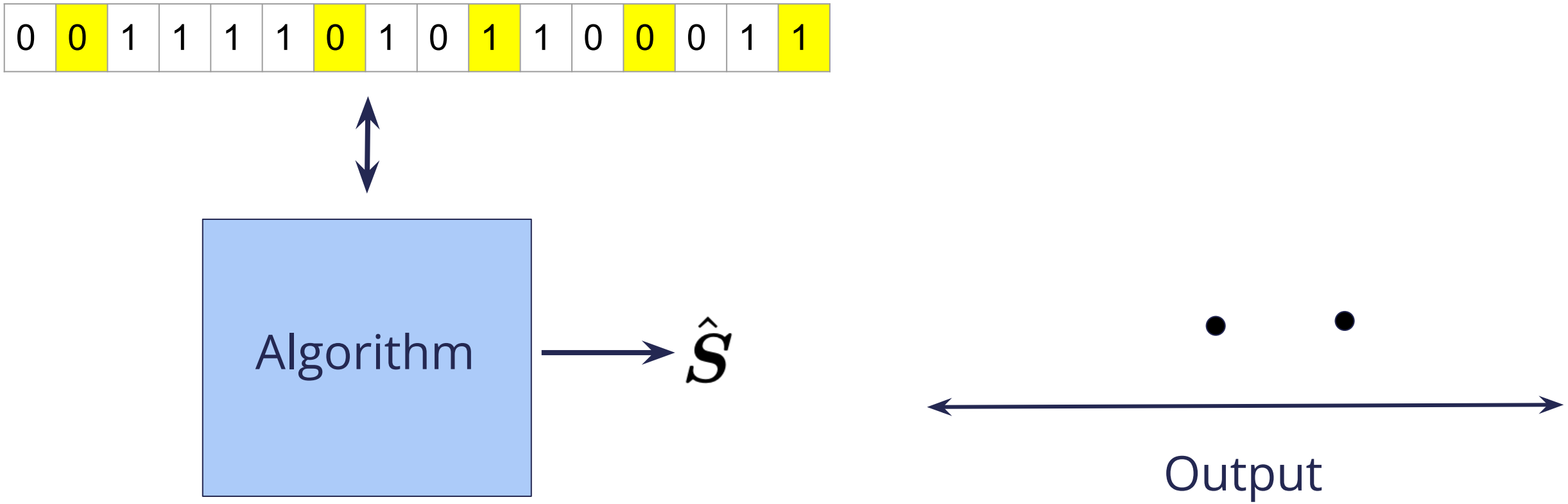

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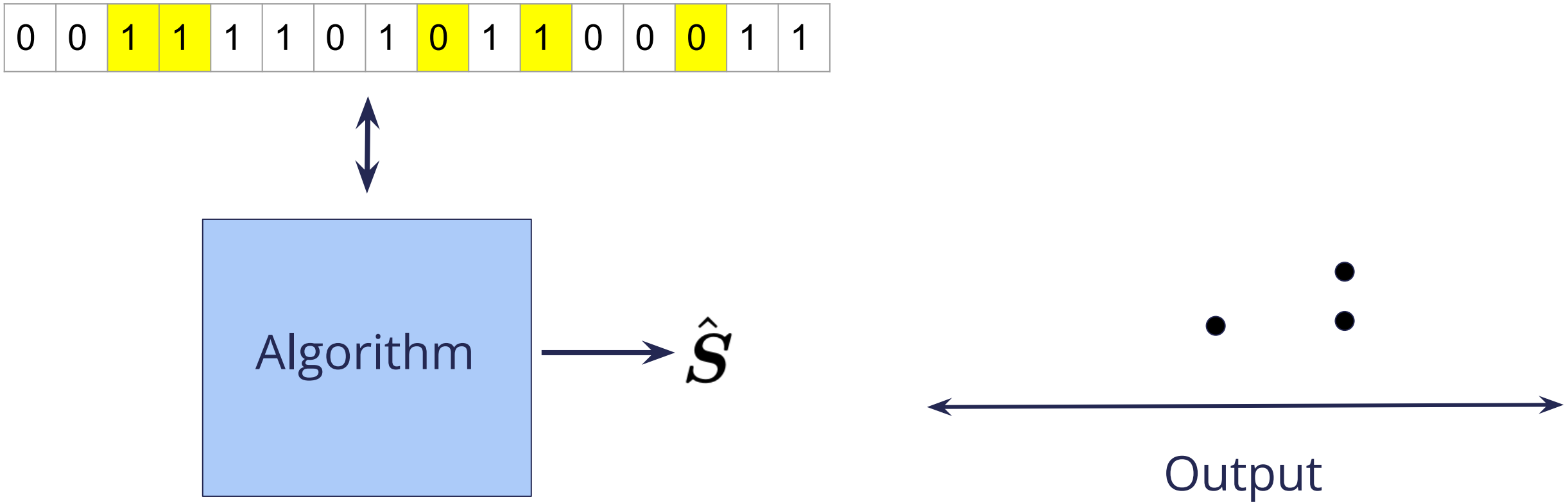

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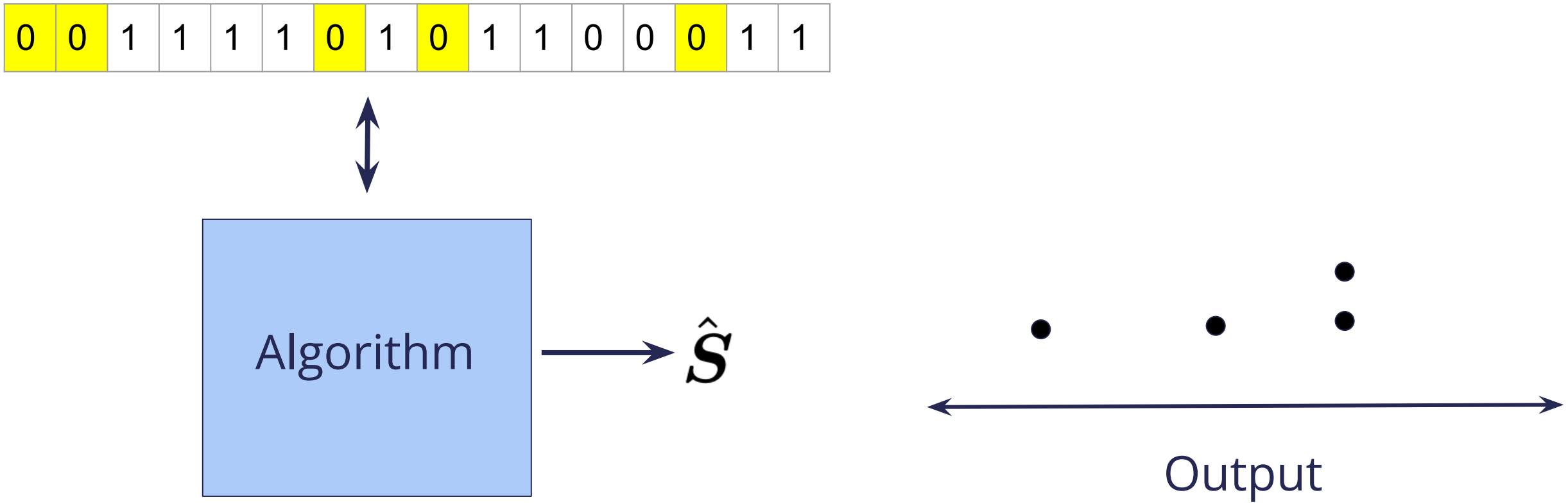
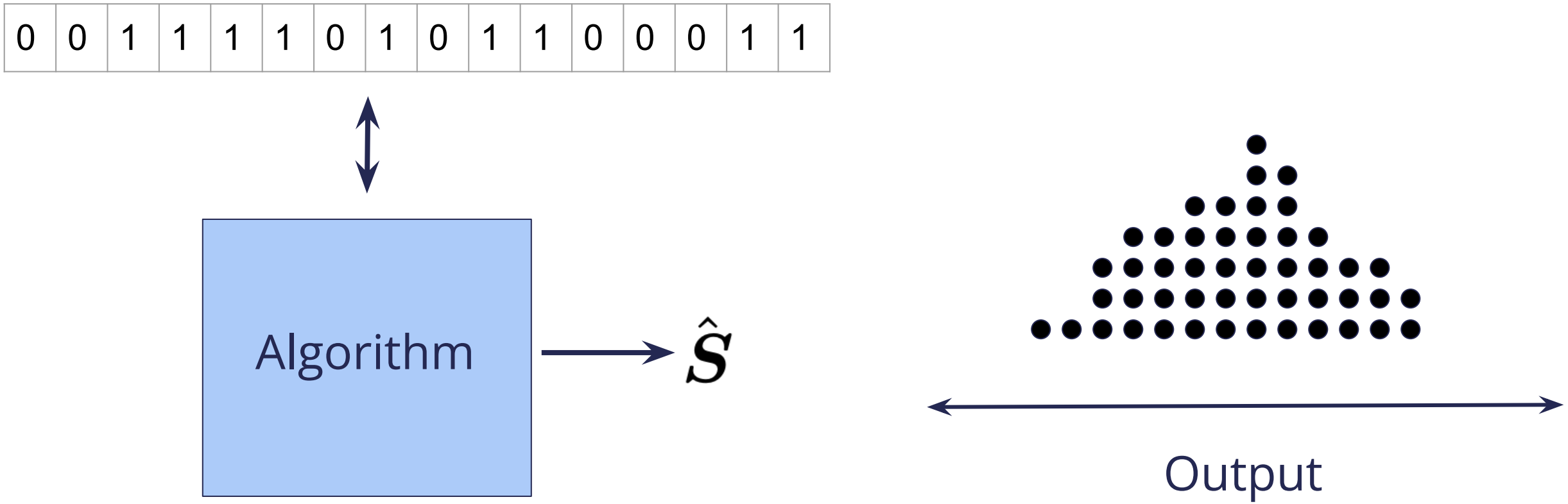
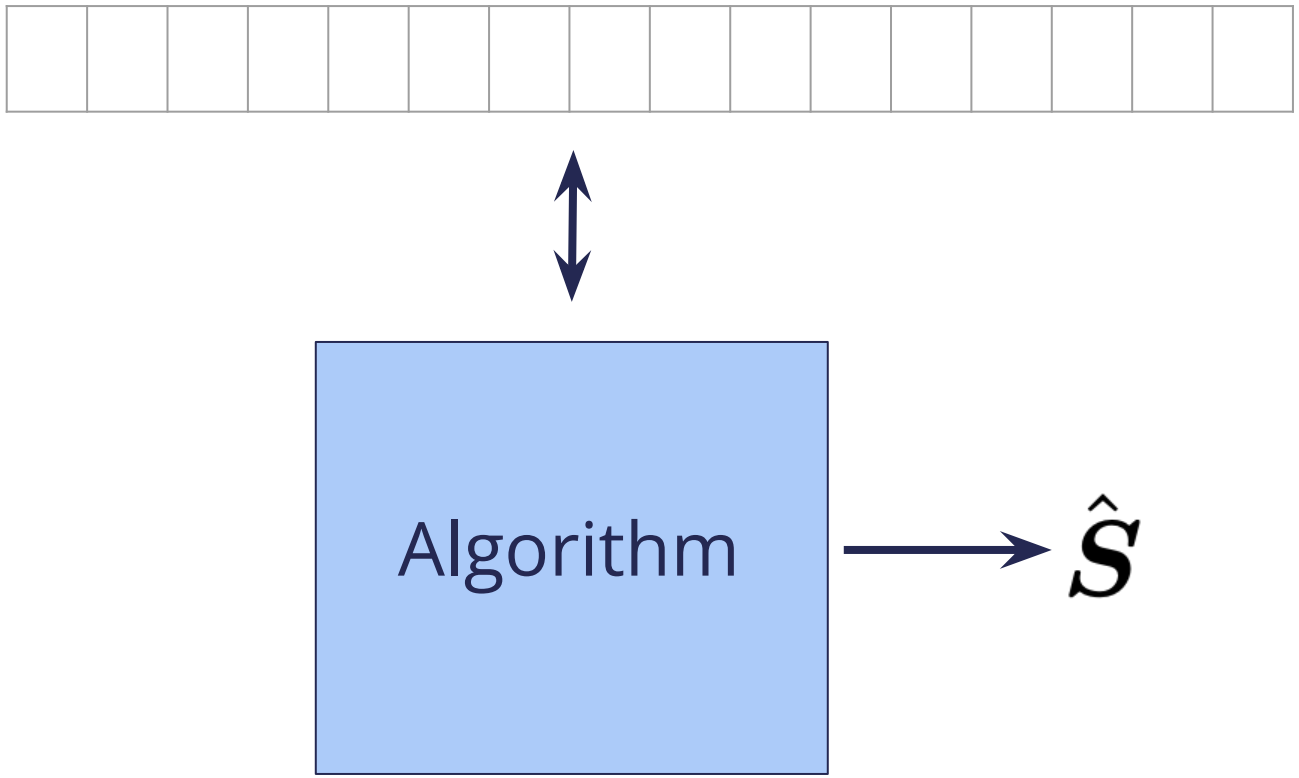


What if the algorithm does not commit to certain queries?



Visualizing a randomized algorithm



- Output of the algorithm is a random variable
- Correctness in terms of properties of the random variable

Pros and Cons of Randomized Algorithms

- Pros:
  - Algorithm doesn't commit to fixed set of queries
  - Cannot construct indistinguishable instances, because we don't know queries

- Cons:
  - Every time we execute algorithm, may get a different answer
  - Hard to debug
  - Algorithm's output is a random variable, must reason probabilistically about guarantees

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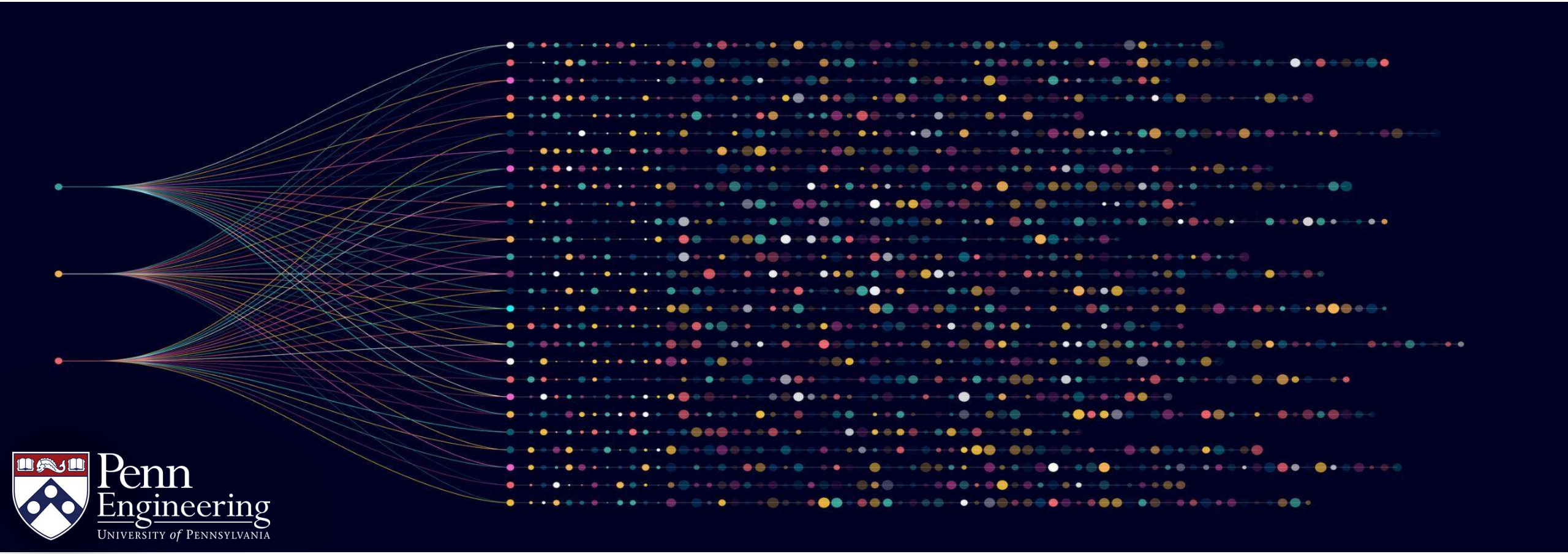
Probabilistic Guarantees



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Objectives: Computing Large Averages

- Probabilistic guarantees for algorithms

- Additive approximations with high probability

Guarantees of Randomized Algorithms

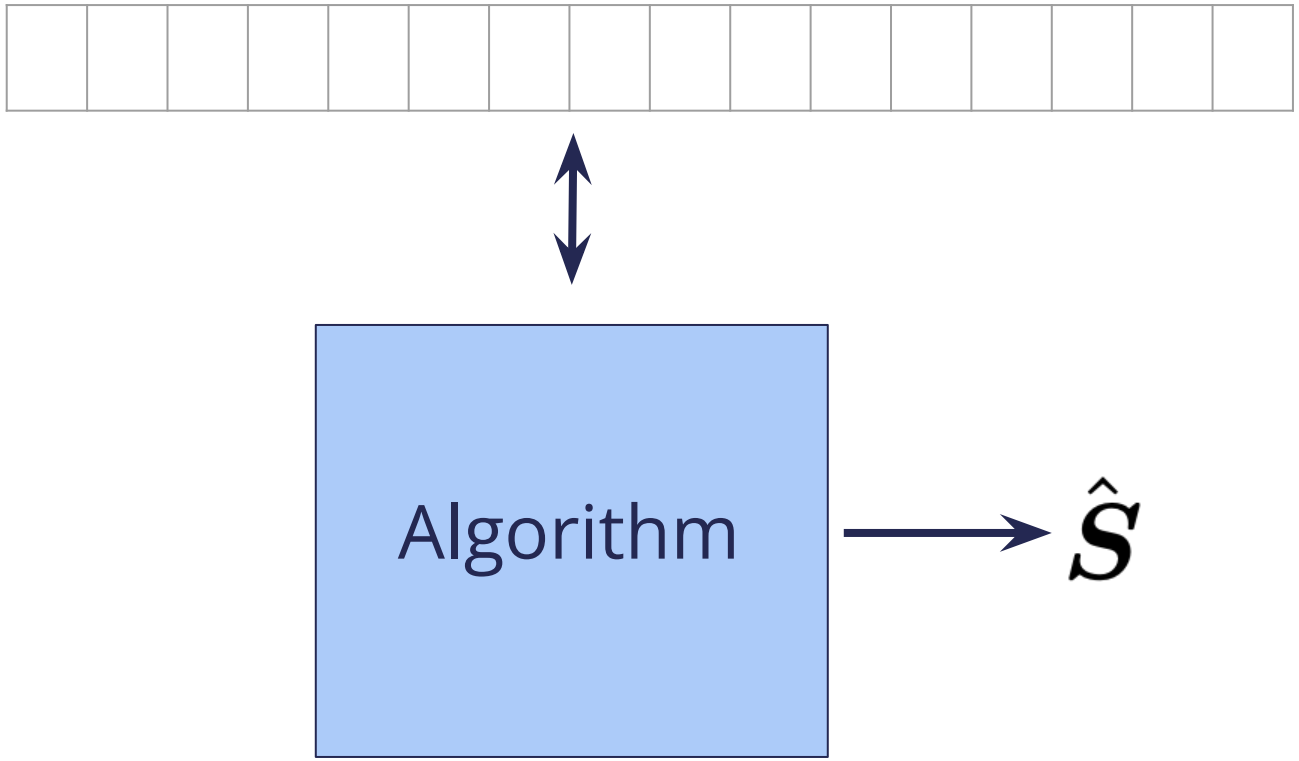
Definition: A randomized algorithm for approximating an average is an algorithm with the following guarantees:

- Input:
  - Query access to $X_1, \ldots, X_n \in \mathbb{R}$
  - Accuracy parameter ϵ
  - Source of randomness
  - Failure probability $\delta \in (0, 1)$
- Guarantee: $$\Pr\left[\left|\hat{\boldsymbol{S}} - \frac{1}{n}\sum_{i=1}^{n} X_i\right| \leq \epsilon\right] \geq 1 - \delta.$$

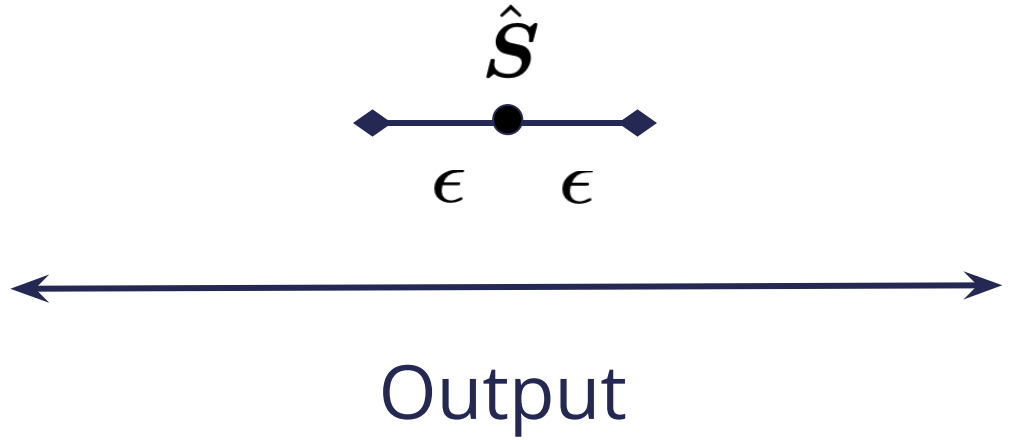
Interpreting results of randomized approximations

- How many users are in Europe?
  - Suppose accuracy is 0.05 and failure probability is 0.01
  - Output of algorithm is 0.3
  - There is a 1% chance algorithm is incorrect. If correct, then between 25% and 35% of users are in Europe.
- How many users posted in the past week?
  - Suppose accuracy is 0.1 and failure probability is 0.05
  - Output of algorithm is 0.9
  - There is a 5% chance algorithm is incorrect. If correct, then between 80% and 100% of users posted in the past week.
- How long (on average) did users spend on the platform today?
  - Suppose accuracy is 30 and failure probability 0.0001
  - Output of algorithm is 120
  - There is a 0.01% chance algorithm is incorrect. If correct, then users spent between 90 and 150 seconds on the platform on average.

Approximate and randomized guarantees: **user perspective**

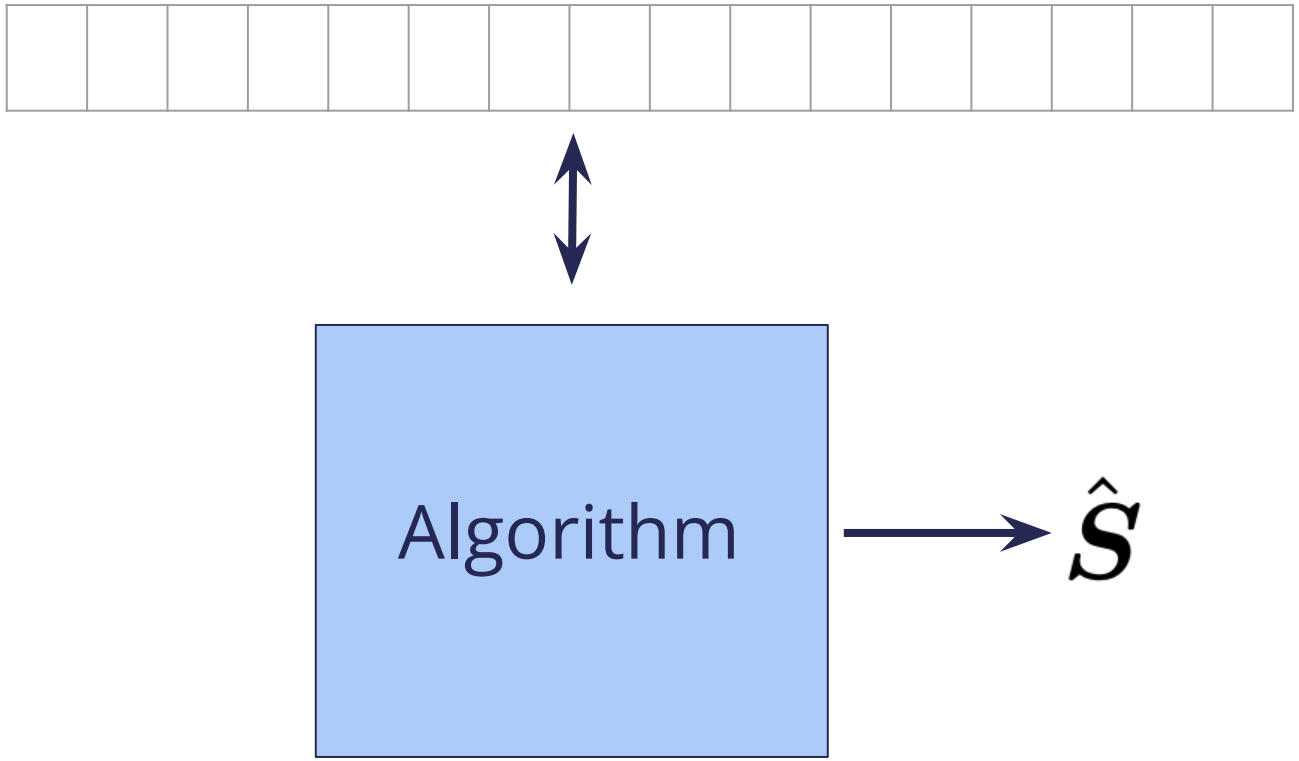


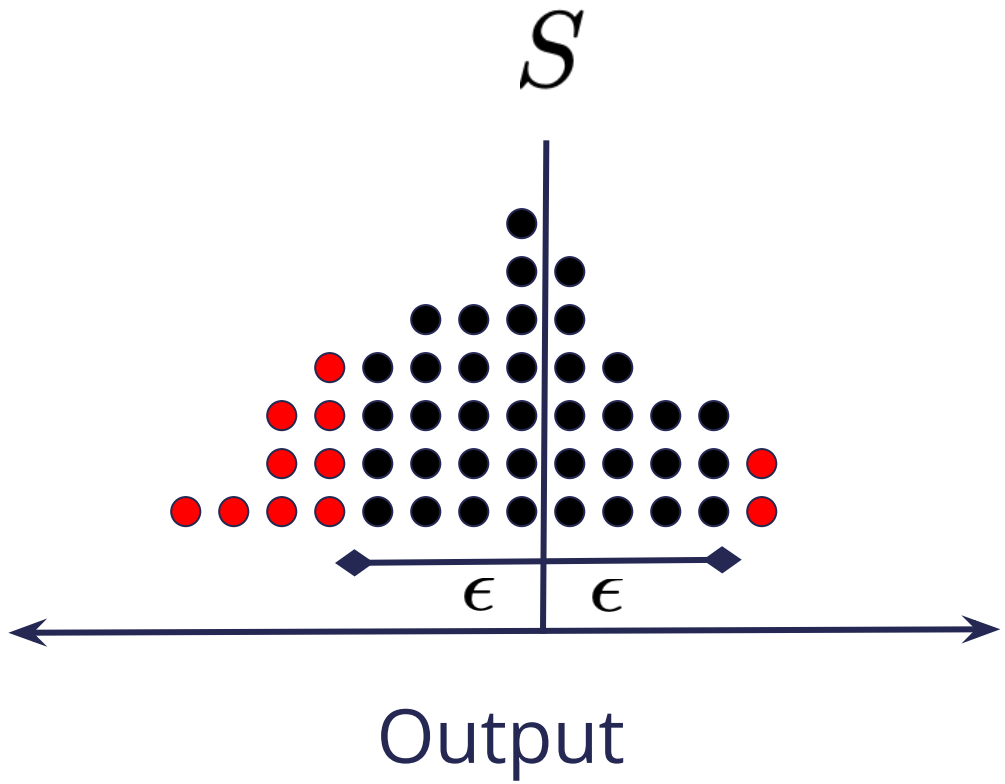
may be wrong with probability δ



ϵ : accuracy δ: failure probability

Approximate and randomized guarantees: **algorithm perspective**





Output

ϵ : accuracy δ: failure probability

Key algorithmic questions:

- Do there exist (more) efficient algorithms when allowing approximation and randomization?

- What is the tradeoff between the number of queries, the accuracy parameter, and failure probability?

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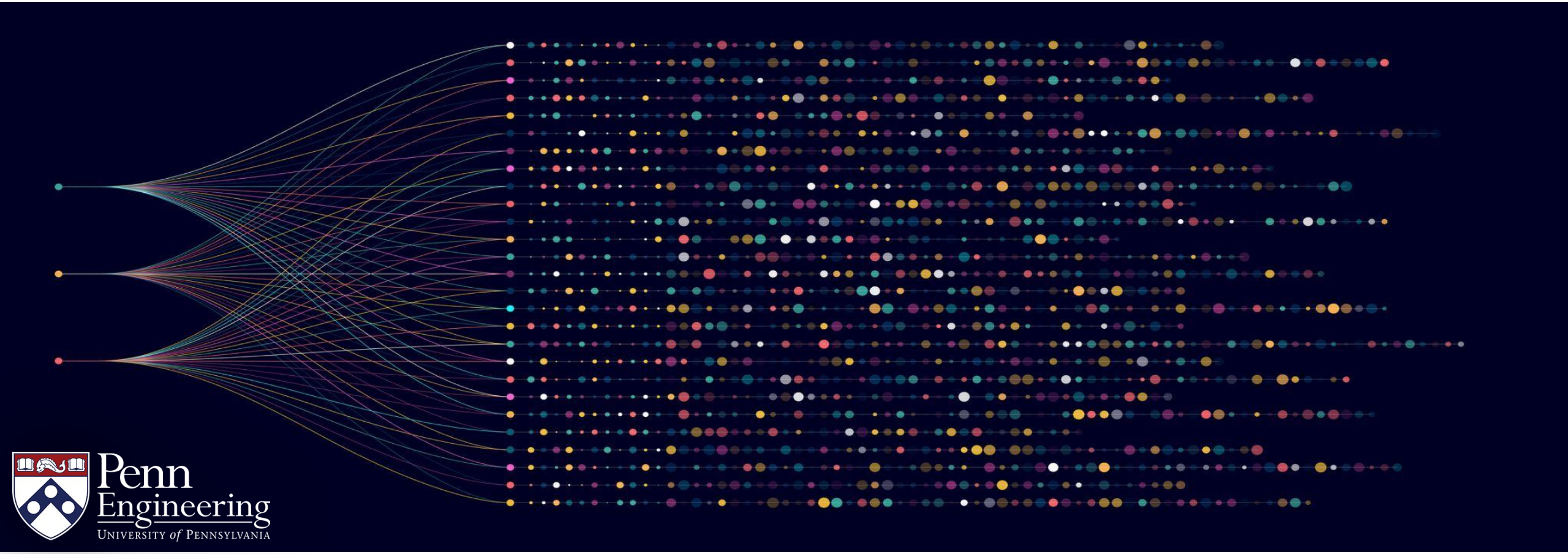
Description of the Algorithm



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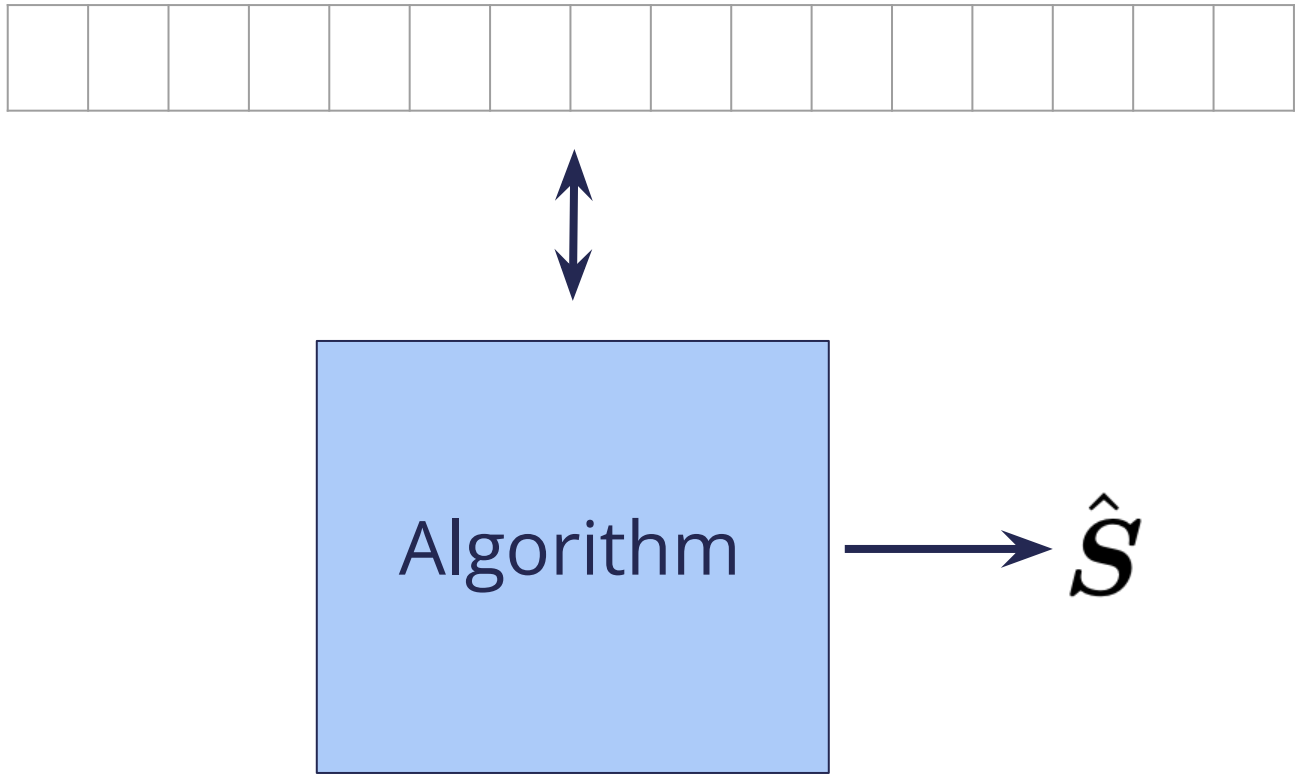
Objectives: Computing Large Averages

- Algorithm for approximating large averages

- Questions for the analysis

Algorithm Description



1. Sample q positions:

$$i_1, \ldots, i_q \sim [n]$$

1. Output average:

$$\hat{S} = \frac{1}{q} \sum_{t=1}^{q} X_{i_t}$$

Understanding the algorithm:

- Queries are each chosen uniformly and independently
- For every $j \in [n]$,

$$\Pr_{\boldsymbol{i}_t \sim [n]} [\boldsymbol{i}_t = j] = \frac{1}{n}$$

1. Sample q positions:

$$\boldsymbol{i}_1, \ldots, \boldsymbol{i}_q \sim [n]$$

1. Output average:

$$\hat{\boldsymbol{S}} = \frac{1}{q} \sum_{t=1}^{q} X_{\boldsymbol{i}_t}$$

Example Execution of Algorithm:

0	0	1	1	1	1	0	1	0	1	1	0	0	0	1	1

$$\boldsymbol{i}_1 =$$

$$\boldsymbol{i}_2 =$$

$$\vdots$$

$$\boldsymbol{i}_q =$$

source of randomness:

1. Sample q positions:

$$\boldsymbol{i}_1, \ldots, \boldsymbol{i}_q \sim [n]$$

1. Output average:

$$\hat{\boldsymbol{S}} = \frac{1}{q} \sum_{t=1}^{q} X_{\boldsymbol{i}_t}$$

Example Execution of Algorithm:

0	0	1	1	1	1	0	1	0	1	1	0	0	0	1	1

$$\boldsymbol{i}_1 = 11$$

$$\boldsymbol{i}_2 = 8$$

$$\vdots$$

$$\boldsymbol{i}_q = 15$$

source of randomness:

1. Sample q positions:

$$\boldsymbol{i}_1, \ldots, \boldsymbol{i}_q \sim [n]$$

1. Output average:

$$\hat{\boldsymbol{S}} = \frac{1}{q} \sum_{t=1}^{q} X_{\boldsymbol{i}_t}$$

Example Execution of Algorithm:

0	0	1	1	1	1	0	1	0	1	1	0	0	0	1	1

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source of randomness:

1. Sample q positions:

$$\boldsymbol{i}_1, \ldots, \boldsymbol{i}_q \sim [n]$$

1. Output average:

$$\hat{\boldsymbol{S}} = \frac{1}{q} \sum_{t=1}^{q} X_{\boldsymbol{i}_t}$$

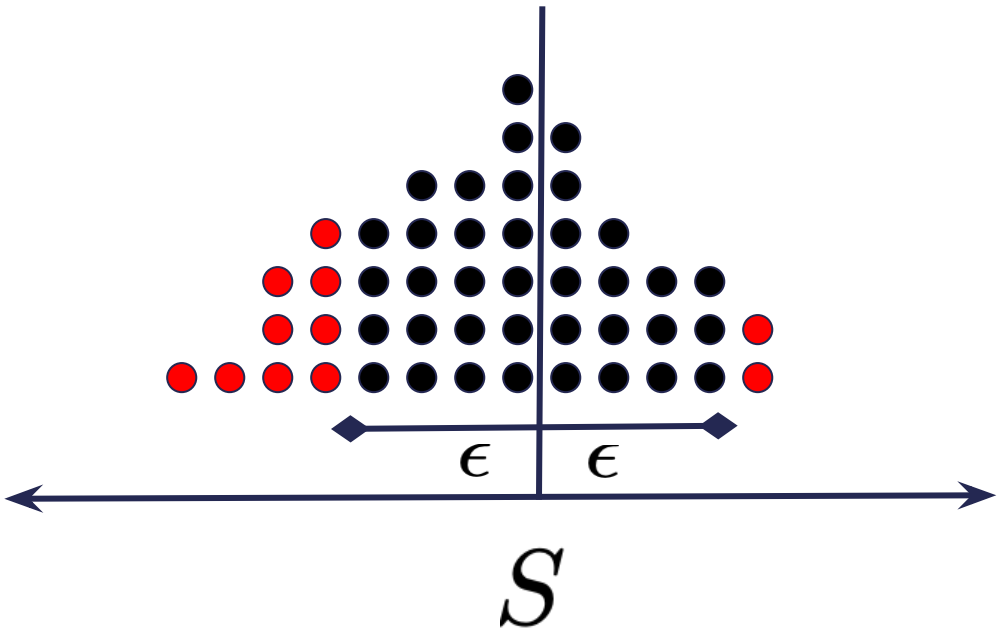
Questions for the Algorithm Analysis:

- How many queries do we need?

$$\Pr_{\boldsymbol{i}_1,\ldots,\boldsymbol{i}_q \sim [n]} \left[\left| \hat{\boldsymbol{S}} - S \right| \leq \epsilon \right] \geq 1 - \delta.$$

- Should depend on ϵ, δ.

Questions for the Algorithm Analysis:



$$\hat{S} = \frac{1}{q}\sum_{t=1}^{q} X_{i_t}$$

$$\Pr_{i_1,\ldots,i_q \sim [n]}\left[\left|\hat{S} - S\right| \leq \epsilon\right] \geq 1 - \delta.$$

ϵ : accuracy δ: failure probability

Toward the Analysis:

- If we took infinitely many queries, would the algorithm's output approach the average?
  - Hint: think of the expectation and the central-limit theorem!

- Suppose yes, with infinitely many queries. What happens with finitely many queries?
  - How close is an outcome to its expectation?
  - Hint: think of the variance and Chebyshev's inequality

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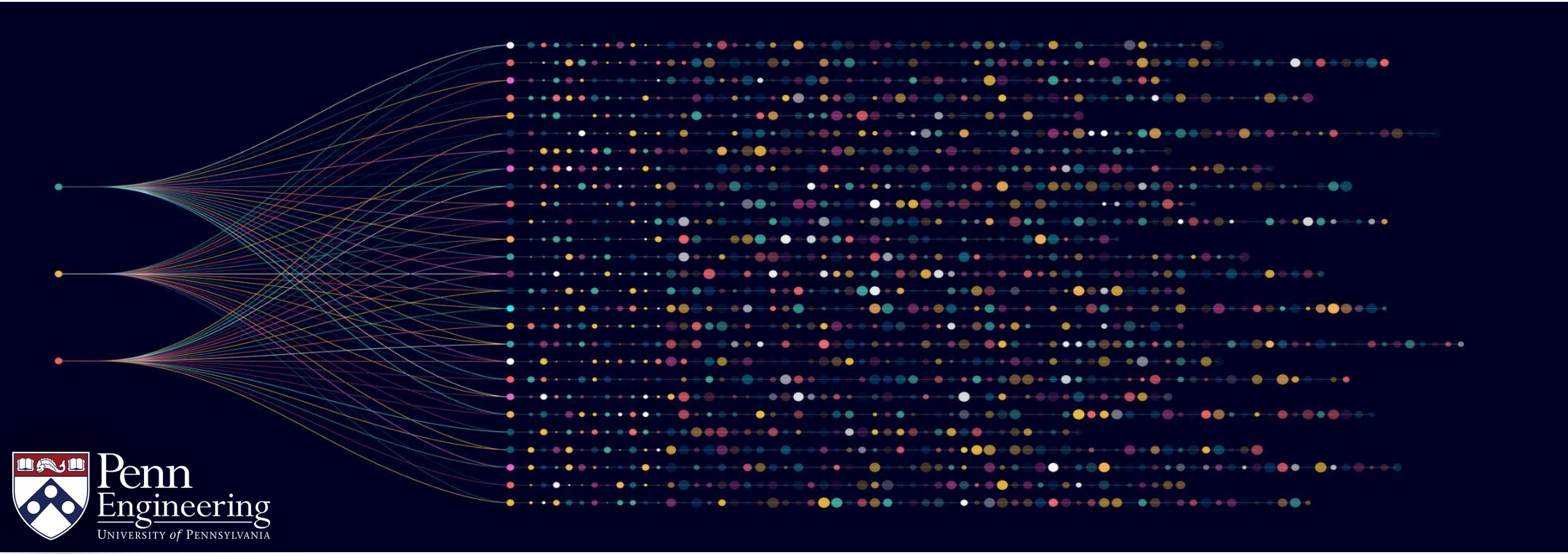
Analysis of the Algorithm



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Objectives: Computing Large Averages

- Analysis of algorithm

- Expectation and Variance analysis

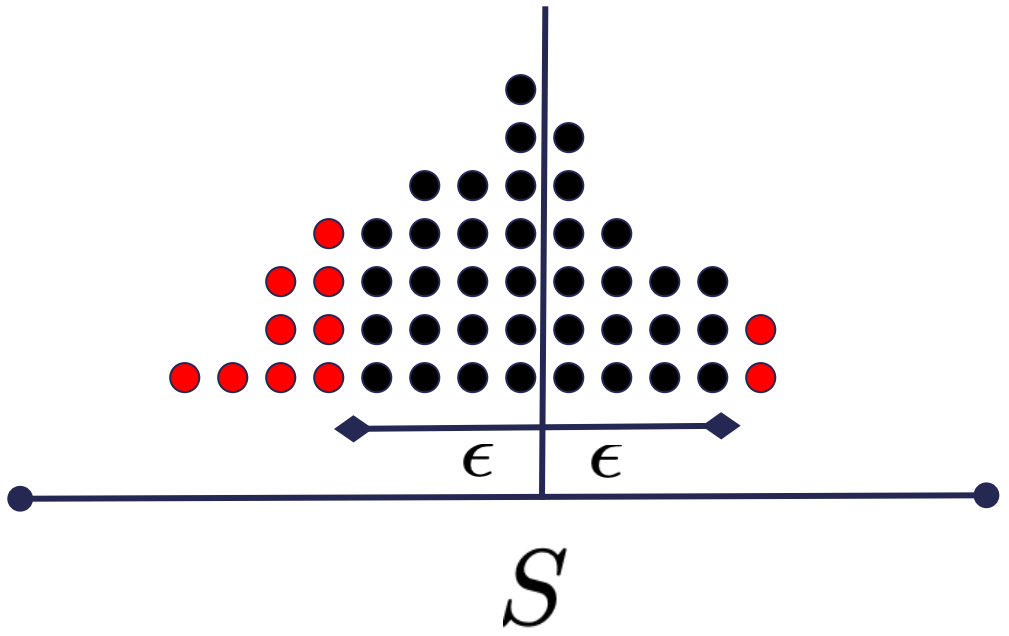
Analysis of the Algorithm

1. Sample q positions:

$$\boldsymbol{i}_1, \ldots, \boldsymbol{i}_q \sim [n]$$

2. Output empirical average:

$$\hat{\boldsymbol{S}} = \frac{1}{q} \sum_{t=1}^{q} X_{\boldsymbol{i}_t}$$



ϵ : accuracy δ : failure probability

Chebyshev's Inequality

- For any random variable $\boldsymbol{Z}$ with expectation $\mathbb{E}[\boldsymbol{Z}] = \mu$

$$\Pr\left[|\boldsymbol{Z} - \mu| \geq \epsilon\right] \leq \frac{\mathrm{Var}[\boldsymbol{Z}]}{\epsilon^2}$$

- Exactly what we need:

$$\Pr\left[\left|\hat{\boldsymbol{S}} - S\right| \geq \epsilon\right] \leq \frac{\mathrm{Var}[\hat{\boldsymbol{S}}]}{\epsilon^2}$$

The Plan:

- Compute the expectation of $\hat{S}$

- Compute an upper bound on $\mathrm{Var}[\hat{S}]$
  - Variance will depend on q

$$\Pr\left[\left|\hat{S} - S\right| \geq \epsilon\right] \leq \frac{\mathrm{Var}[\hat{S}]}{\epsilon^2}$$

Analysis via Chebyshev's Inequality

- Simplifying Expectation of $\hat{S}$:

$$\mathop{\mathbb{E}}_{i_1,\ldots,i_q \sim [n]} \left[\hat{S}\right] = \mathop{\mathbb{E}}_{i_1,\ldots,i_q \sim [n]} \left[\frac{1}{q}\sum_{t=1}^{q} X_{i_t}\right]$$

$$= \frac{1}{q}\sum_{t=1}^{q} \mathop{\mathbb{E}}_{i_t \sim [n]} \left[X_{i_t}\right]$$

Analysis via Chebyshev's Inequality

- Simplifying Expectation of $\hat{S}$:

$$\mathop{\mathbb{E}}_{i_1,\ldots,i_q \sim [n]} \left[\hat{S}\right] = \frac{1}{q} \sum_{t=1}^{q} \mathop{\mathbb{E}}_{i_t \sim [n]} \left[X_{i_t}\right]$$

- Expectation of each term:

$$\mathop{\mathbb{E}}_{i_t \sim [n]} \left[X_{i_t}\right] = \sum_{j=1}^{n} \Pr[i_t = j] \cdot X_j = \frac{1}{n} \sum_{j=1}^{n} X_j = S.$$

Chebyshev's Inequality

- For any random variable $\boldsymbol{Z}$ with expectation $\mathbb{E}[\boldsymbol{Z}] = \mu$

$$\Pr\left[|\boldsymbol{Z} - \mu| \geq \epsilon\right] \leq \frac{\mathrm{Var}[\boldsymbol{Z}]}{\epsilon^2}$$

- Exactly what we need:

$$\Pr\left[\left|\hat{\boldsymbol{S}} - S\right| \geq \epsilon\right] \leq \frac{\mathrm{Var}[\hat{\boldsymbol{S}}]}{\epsilon^2}$$

Task:

- Compute an upper bound on $\mathrm{Var}[\hat{S}]$ as a function of q
  - As q increases, variance will decrease

- Set q large enough so that

$$\Pr\left[\left|\hat{S} - S\right| \geq \epsilon\right] \leq \frac{\mathrm{Var}[\hat{S}]}{\epsilon^2} \leq \delta.$$

Variance:

- Simplifying Variance:

$$\mathrm{Var}[\hat{\boldsymbol{S}}] = \mathrm{Var}\left[\frac{1}{q}\sum_{t=1}^{q} X_{\boldsymbol{i}_t}\right]$$

$$= \frac{1}{q^2}\sum_{t=1}^{q} \mathrm{Var}\left[X_{\boldsymbol{i}_t}\right]$$

Variance:

- Simplifying Variance:

$$\mathrm{Var}[\hat{\boldsymbol{S}}] = \frac{1}{q^2} \sum_{t=1}^{q} \mathrm{Var}\left[X_{\boldsymbol{i}_t}\right]$$

- Variance of single term:

$$\begin{aligned} \mathrm{Var}\left[X_{\boldsymbol{i}_t}\right] &= \mathbb{E}\left[\left(X_{\boldsymbol{i}_t} - S\right)^2\right] \\ &\leq 1. \end{aligned}$$

Variance:

- Simplifying Variance:

$$\mathrm{Var}[\hat{\boldsymbol{S}}] = \frac{1}{q^2}\sum_{t=1}^{q}\mathrm{Var}\left[X_{\boldsymbol{i}_t}\right] \leq \frac{1}{q}$$

- Variance of single term:

$$\begin{aligned}\mathrm{Var}\left[X_{\boldsymbol{i}_t}\right] &= \mathbb{E}\left[\left(X_{\boldsymbol{i}_t} - S\right)^2\right] \\ &\leq 1.\end{aligned}$$

Chebyshev's inequality:

- Substituting bound on variance:

$$\Pr\left[\left|\hat{\boldsymbol{S}} - S\right| \geq \epsilon\right] \leq \frac{\mathrm{Var}[\hat{\boldsymbol{S}}]}{\epsilon^2} \leq \frac{1}{q\epsilon^2}.$$

- Set $q = \dfrac{1}{\delta\epsilon^2}$:

$$\Pr\left[\left|\hat{\boldsymbol{S}} - S\right| \geq \epsilon\right] \leq \delta$$

Complete Description of Algorithm:

1. Sample $q = \frac{1}{\delta \epsilon^2}$ positions:

$$\boldsymbol{i}_1, \ldots, \boldsymbol{i}_q \sim [n]$$

2. Output empirical average:

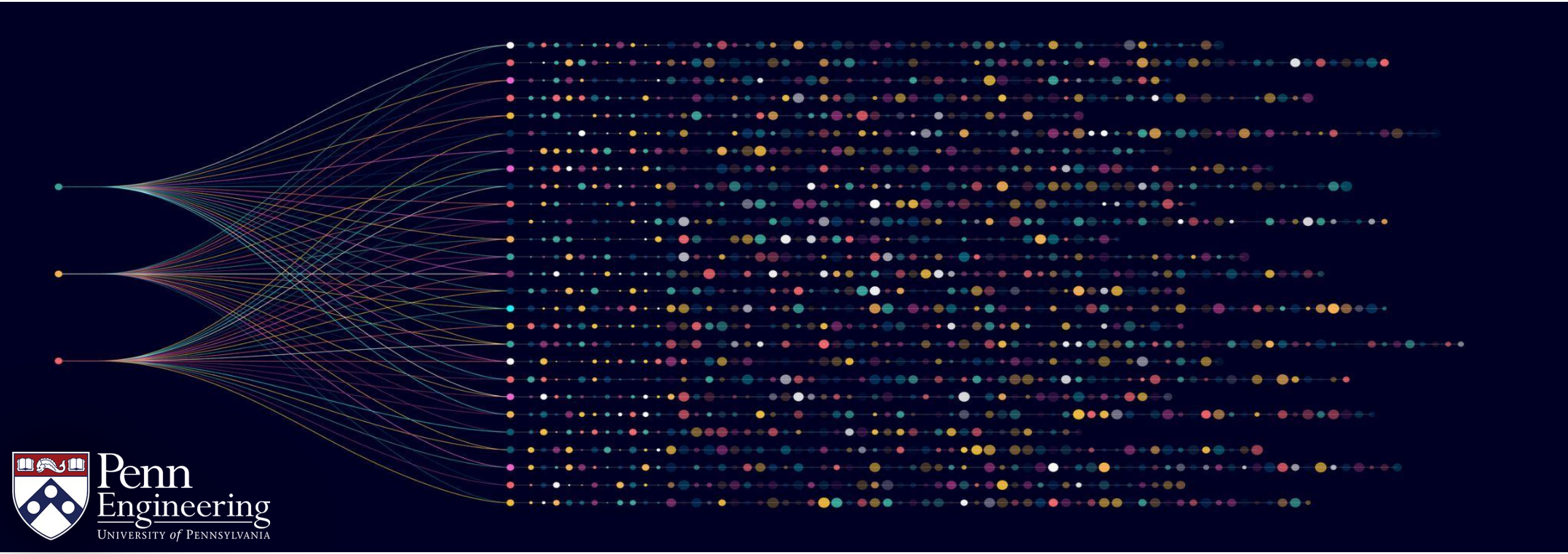
$$\hat{\boldsymbol{S}} = \frac{1}{q} \sum_{t=1}^{q} X_{\boldsymbol{i}_t}$$

- Completely independent of the size of the data!
- Is it optimal?

Algorithms for Big Data

Anindya De Erik Waingarten

Example Setting of Parameters



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Objectives: Computing Large Averages

- Example setting of parameters

- Optimality with respect to δ

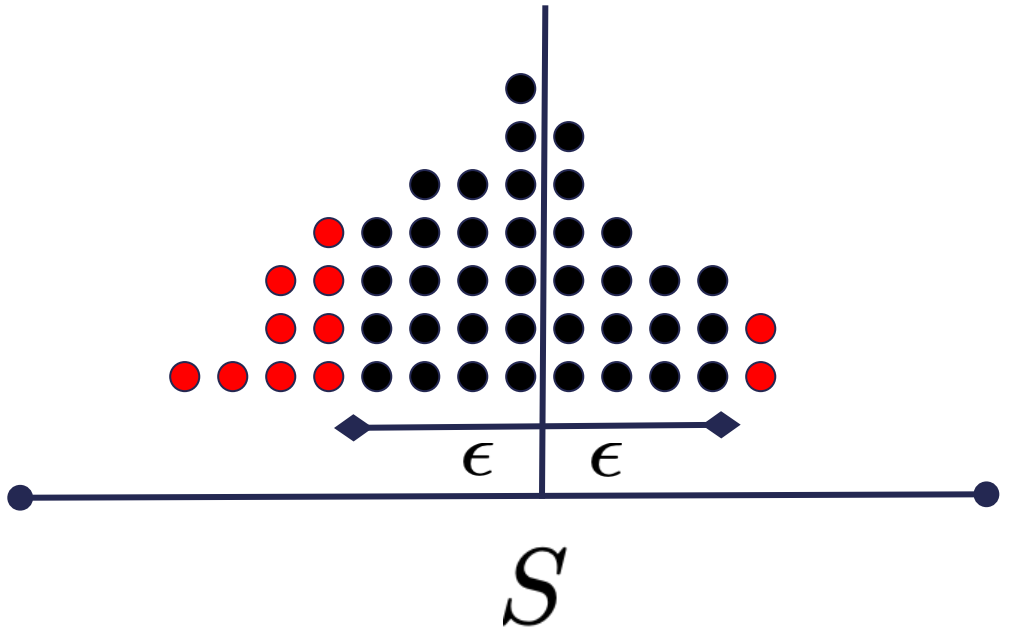
Analysis of the Algorithm

1. Sample $q = \dfrac{1}{\delta\epsilon^2}$ positions:

$$\boldsymbol{i}_1, \ldots, \boldsymbol{i}_q \sim [n]$$

2. Output empirical average:

$$\hat{\boldsymbol{S}} = \frac{1}{q}\sum_{t=1}^{q} X_{\boldsymbol{i}_t}$$



$$\Pr\left[\left|\hat{\boldsymbol{S}} - S\right| \geq \epsilon\right] \leq \delta$$

How many users are in Europe?

- Total number of Facebook users: ~3 billion

- For accuracy parameter 0.01, and failure probability 0.1, it suffices to consider 100,000 queries.
  - That's only 0.003% of the dataset!

- The output is correct 90% of the time. If it is correct, the fraction of European users is $\hat{S}$ up to plus or minus 0.01.

How many users are in Europe?

- Total number of Facebook users: ~3 billion

- For accuracy parameter ~~0.01~~ 0.05, and failure probability 0.1, it suffices to consider ~~100,000~~ 4,000 queries.

- The output is correct 90% of the time. If it is correct, the fraction of European users is $\hat{S}$ up to plus or minus ~~0.01~~ 0.05

Is it optimal?

- Two parameters: accuracy and failure probability
- Query complexity: $q = \frac{1}{\delta\epsilon^2}$
- Dependence on ϵ is optimal
- Dependence on δ can be improved $O\left(\frac{\log(1/\delta)}{\epsilon^2}\right)$

Algorithms for Big Data

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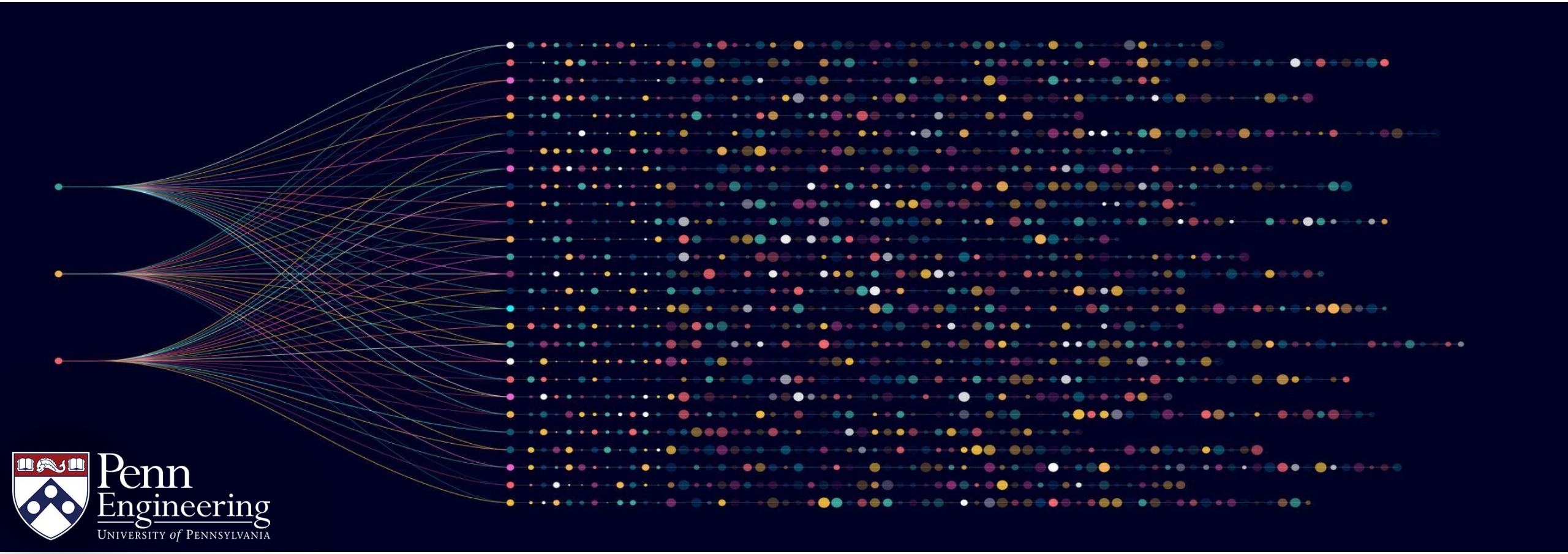
High Probability Guarantees



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Objectives: Computing Large Averages

- Application of Chernoff/Hoeffding inequality

Chebyshev vs. Chernoff/Hoeffding

- Chebyshev Inequality: for random variable Z with $\mathbb{E}[Z] = \mu$

$$\Pr[|Z - \mu| \geq \epsilon] \leq \frac{\mathrm{Var}[Z]}{\epsilon^2}$$

- Chernoff/Hoeffding Inequality
  - If Z is a sum of q independent random variables between a and b:

$$\Pr[|Z - \mu| \geq \epsilon] \leq 2e^{-2\epsilon^2/(q(b-a)^2)}$$

Application of Chernoff/Hoeffding

- Chernoff/Hoeffding Inequality:

$$\Pr\left[\left|\boldsymbol{Z} - \mu\right| \geq \epsilon\right] \leq 2e^{-2\epsilon^2/(q(b-a)^2)}$$

- Our setting: $\hat{\boldsymbol{S}} = \frac{1}{q}\sum_{t=1}^{q} X_{\boldsymbol{i}_t} \qquad b = 1/q \qquad a = 0$

$$\Pr\left[\left|\hat{\boldsymbol{S}} - S\right| \geq \epsilon\right] \leq 2e^{-2\epsilon^2/(q(1/q)^2)} = 2e^{-2\epsilon^2 q}$$

- Setting $q = \frac{\log(2/\delta)}{2\epsilon^2}$ gives desired bound!

Complete Description of Algorithm:

1. Sample $q = \dfrac{\log(2/\delta)}{2\epsilon^2}$ positions:

$$\boldsymbol{i}_1, \ldots, \boldsymbol{i}_q \sim [n]$$

2. Output empirical average:

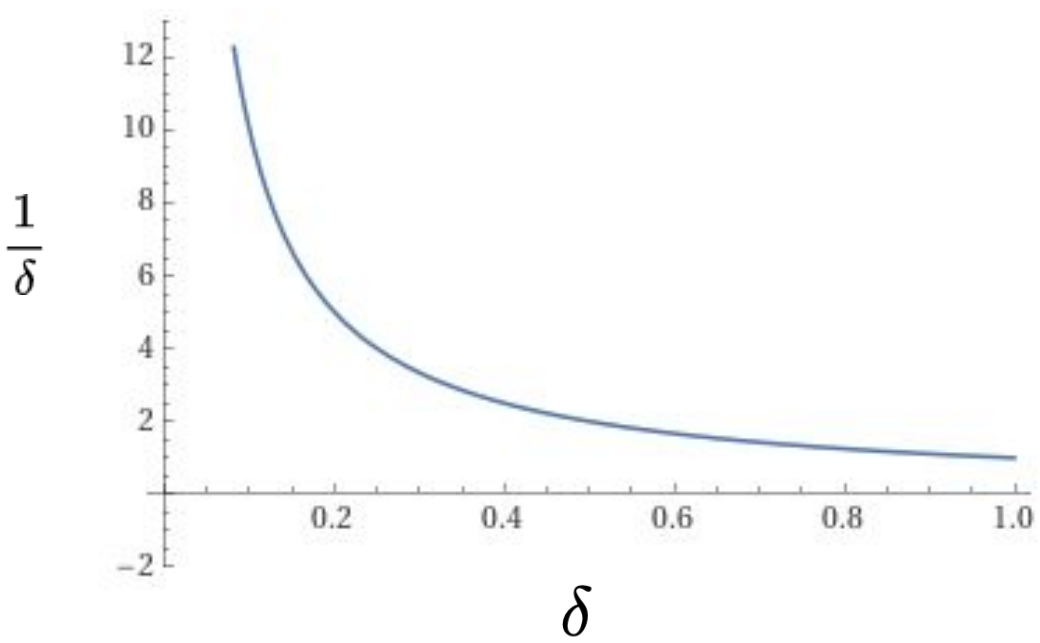
$$\hat{\boldsymbol{S}} = \frac{1}{q} \sum_{t=1}^{q} X_{\boldsymbol{i}_t}$$

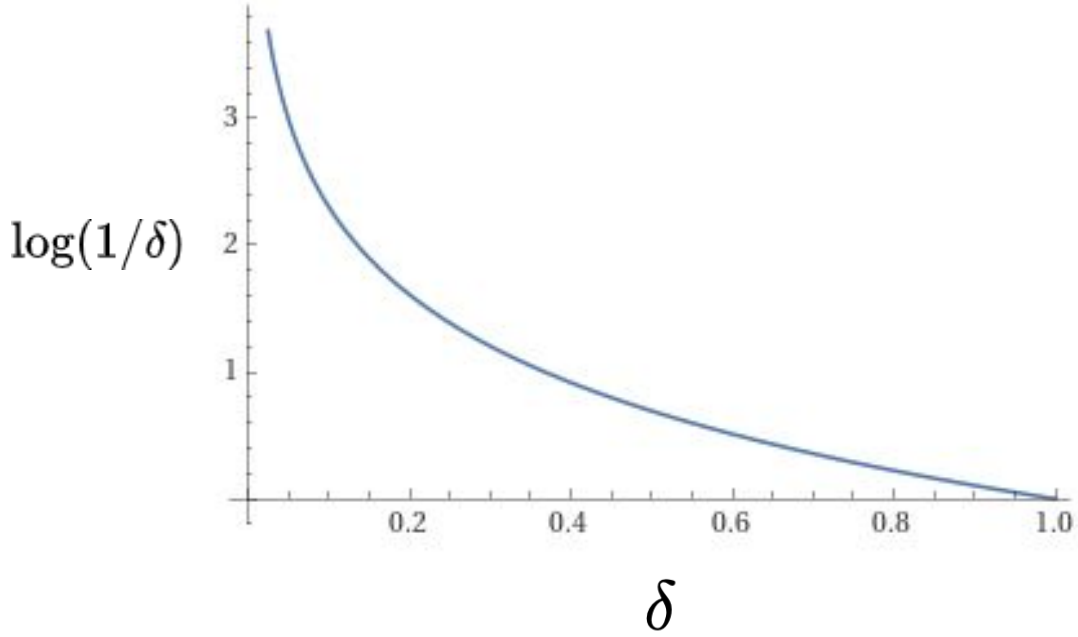
- Asymptotically optimal dependence on accuracy and failure probability

$$O\left(\frac{1}{\delta\epsilon^2}\right) \quad \text{vs.}$$

$$O\left(\frac{\log(1/\delta)}{\epsilon^2}\right)$$

Inverse vs logarithmic dependence





- Logarithmic dependence is always better:
  - For failure probability 0.01 – inverse 100, log (base 2) is 4.6

- Much more pronounced when failure probability is really tiny

Take-aways:

- Two parameters: accuracy and failure probability

$$q = O\left(\frac{\log(1/\delta)}{\epsilon^2}\right)$$

- Easy to make failure probability really *tiny*, but hard to make accuracy really *tiny*

ϵ

δ	0.1	0.01	0.001
0.1	231	23K	2.3M
0.01	460	46K	4.6M
0.001	690	69K	6.9M

Why should you care?

- Relatively cheap to make randomized algorithm succeed with very high probability

- If algorithm fails with probability at most 2^{-300}, it will essentially never fail
  - with inverse relationship: intractable.
  - with logarithmic relationship: tractable.

- Very important when using repeated estimations

Algorithms for Big Data

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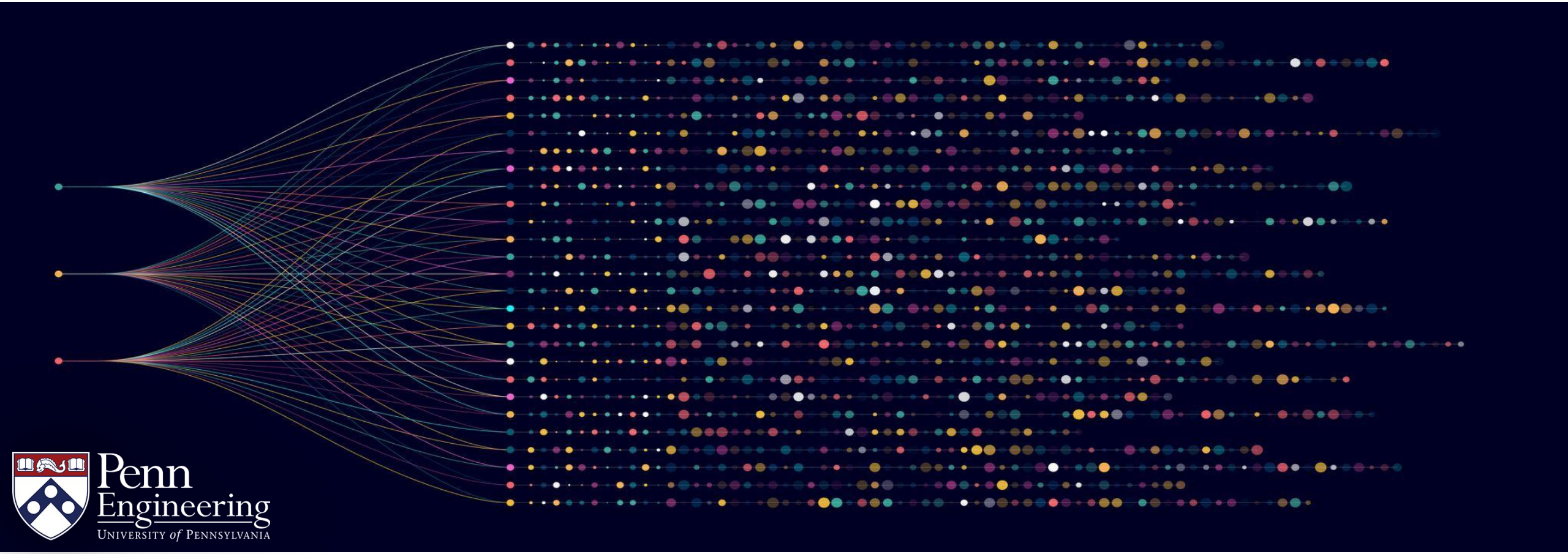
Small Summaries from Sampling



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Objectives: Computing Large Averages

- Application: small summaries via random sampling

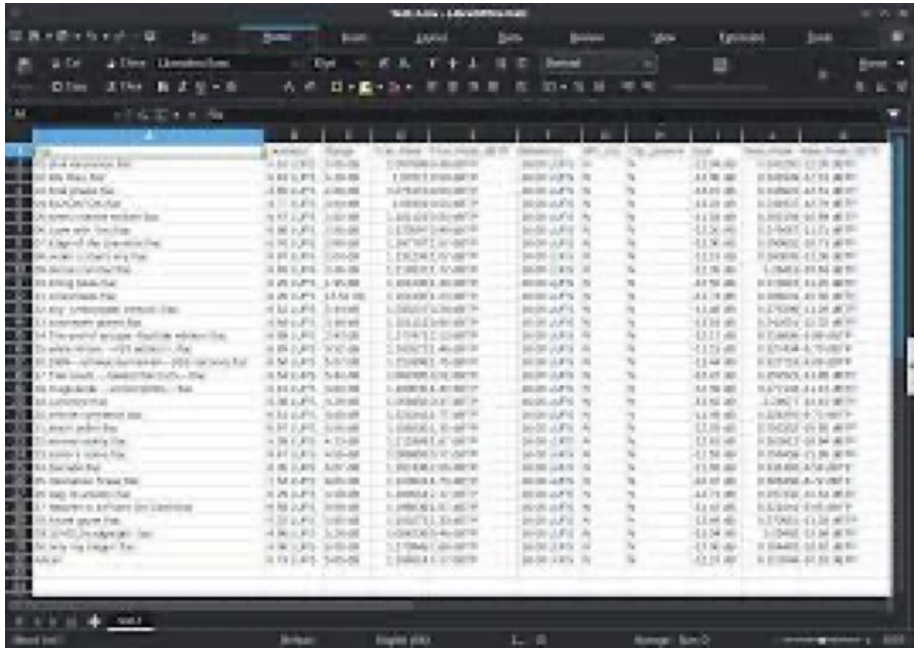
- Small failure probability critically important

Motivating Scenario:

- Users have many attributes
  - approx. statistics on a lot of different user behavior

- How many European? How many are German? British? How many posted today? Yesterday? Logged in twice in a week? ...

- Unclear which questions will arise

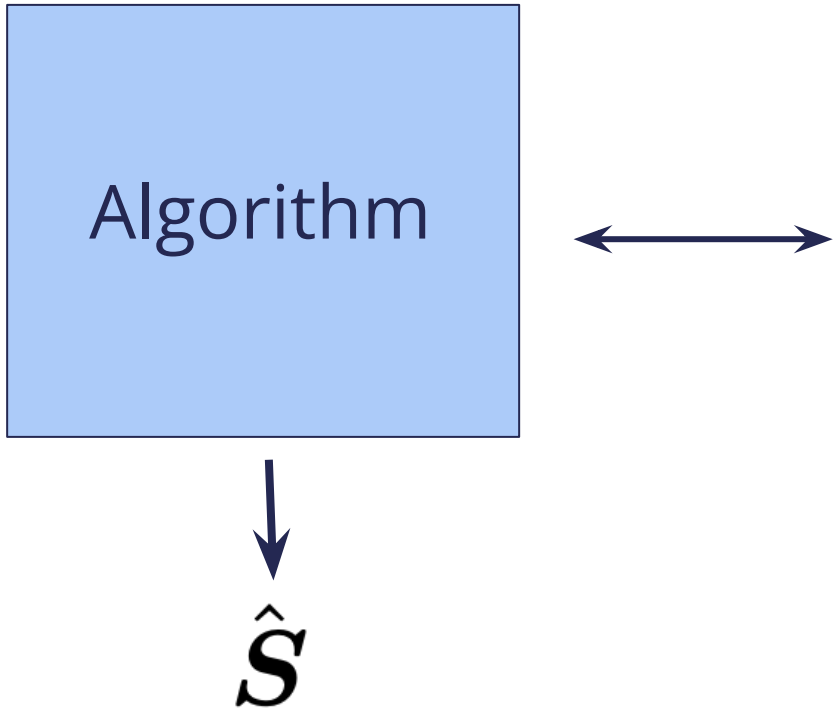


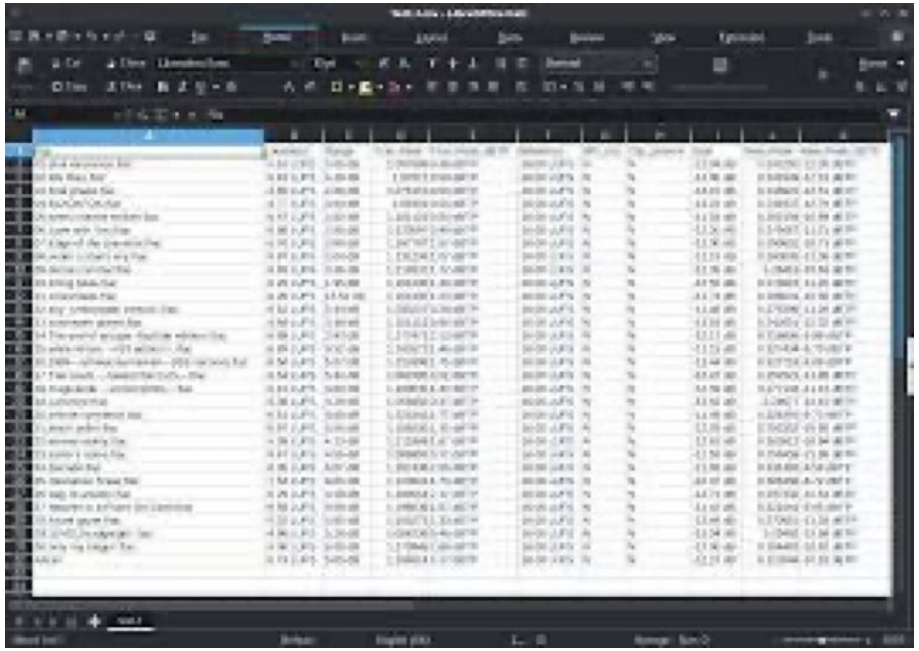
Rows are users,

columns are attributes

Motivating Scenario:

For each question:





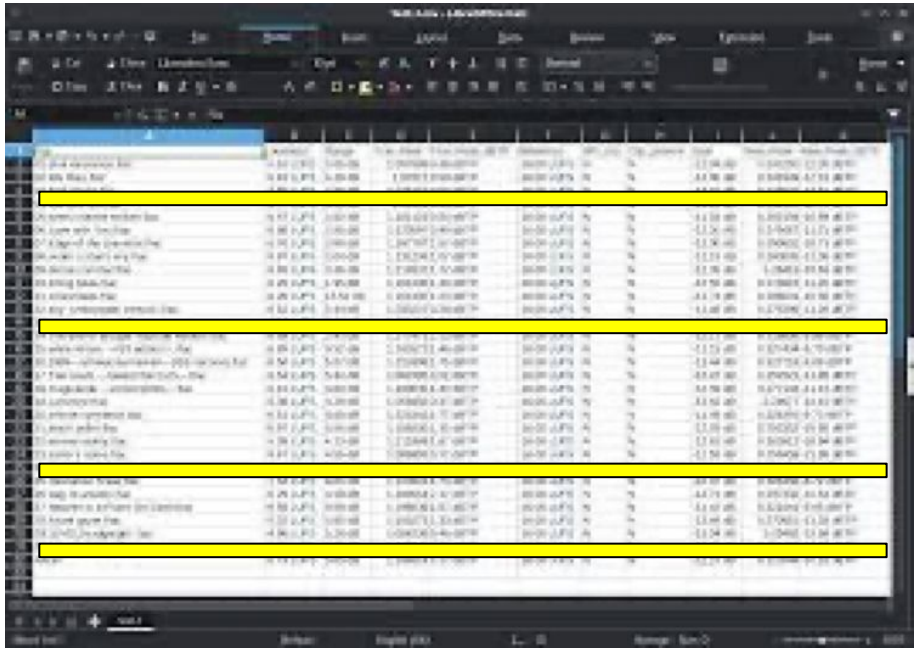
Rows are users,

columns are attributes

Motivating Scenario:

- Each execution generates a random sample of users

- If repeated for M questions, would be using M times q queries

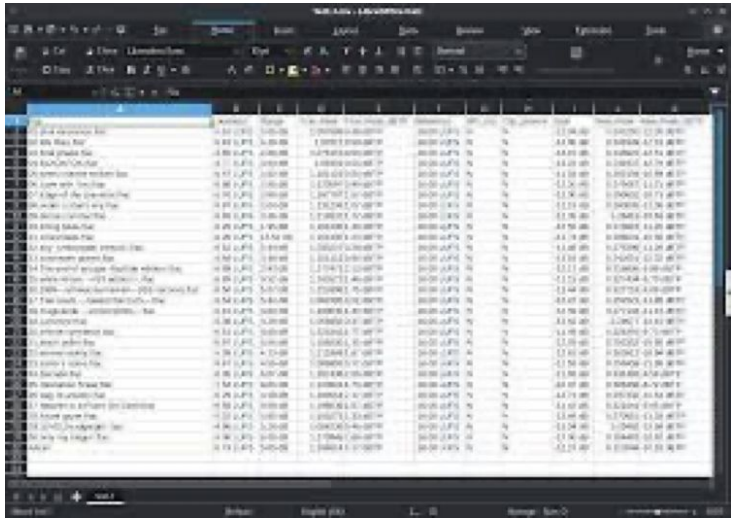


Rows are users,

columns are attributes

Motivating Scenario: Is it possible to…?

- Generate a *single* set of samples



- For each question, use the *same* set of samples to execute the algorithm

$$\boldsymbol{i}_1, \ldots, \boldsymbol{i}_q \sim [n]$$

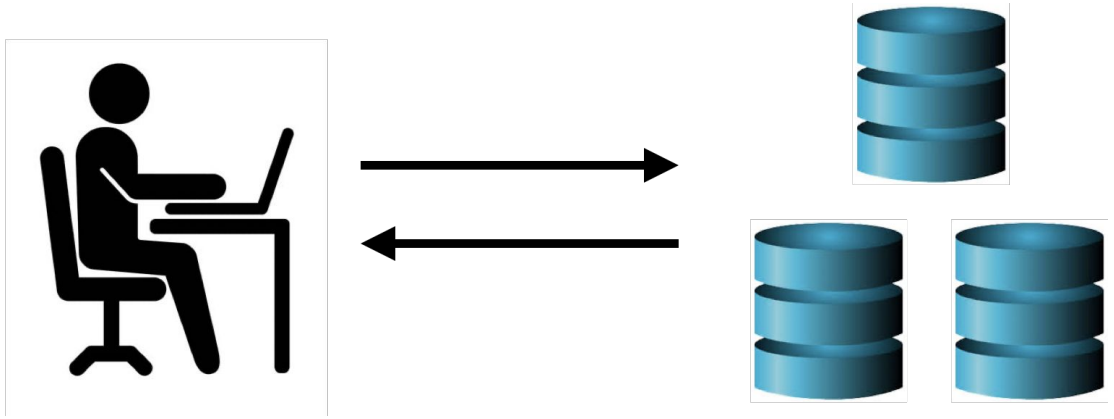
q

Benefits:

- Ideally, fewer samples than one batch per question
- Samples can be pre-generated and used multiple times
- Fewer interactions with central server (also be preemptively done)

Mathematical Formalism:

- Domain $\mathcal{X}$ of possible objects
  - ex. possible space of users
- There is a dataset $P \subset \mathcal{X}$ of n objects
  - ex. possible users
- Set of functions $\mathcal{F}$ mapping domain to [0,1]



$$f \in \mathcal{F} \quad : \quad f : \mathcal{X} \rightarrow [0, 1]$$

Example: Country of Origin for Users

- All possible users $\mathcal{X}$.
- Our dataset of users $P \subset \mathcal{X}$.
- Collection of 195 functions:

$$\mathcal{F} = \{f_{\text{USA}}, f_{\text{Canada}}, f_{\text{Egypt}}, \dots\}$$

$$f_{\text{USA}}(x) = \begin{cases} 1 & x \text{ from USA} \\ 0 & \text{otherwise} \end{cases}$$

Main Result:

Theorem: For any collection of functions $\mathcal{F}$, any dataset P, and any accuracy parameter ϵ and failure probability δ.

Letting

$$q = \frac{\log(2|\mathcal{F}|/\delta)}{2\epsilon^2}$$

A summary of q random samples can ϵ-approximate every function $f \in \mathcal{F}$ with probability at least $1 - \delta$

Applying the Theorem:

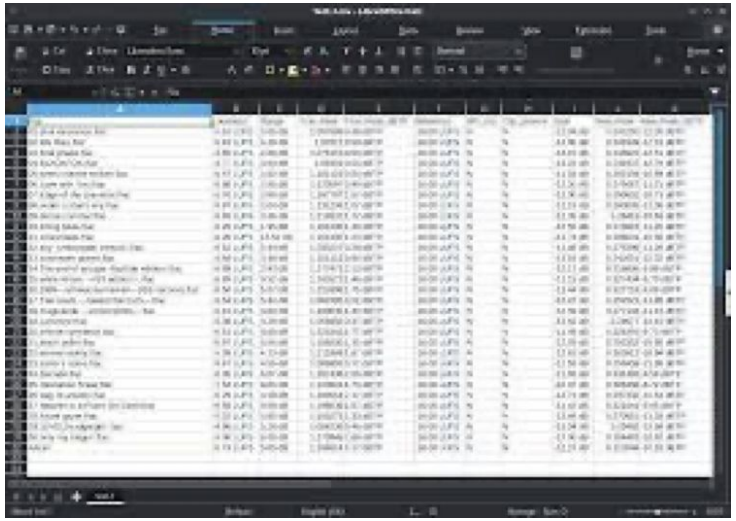
- Fix collection of functions $\mathcal{F}$

- Estimate statistic:

$$\frac{1}{q}\sum_{t=1}^{q} f(x_{i_t})$$

- Number of samples:

$$O(\log(|\mathcal{F}|/\delta)/\epsilon^2)$$



$$i_1, \ldots, i_q \sim [n]$$

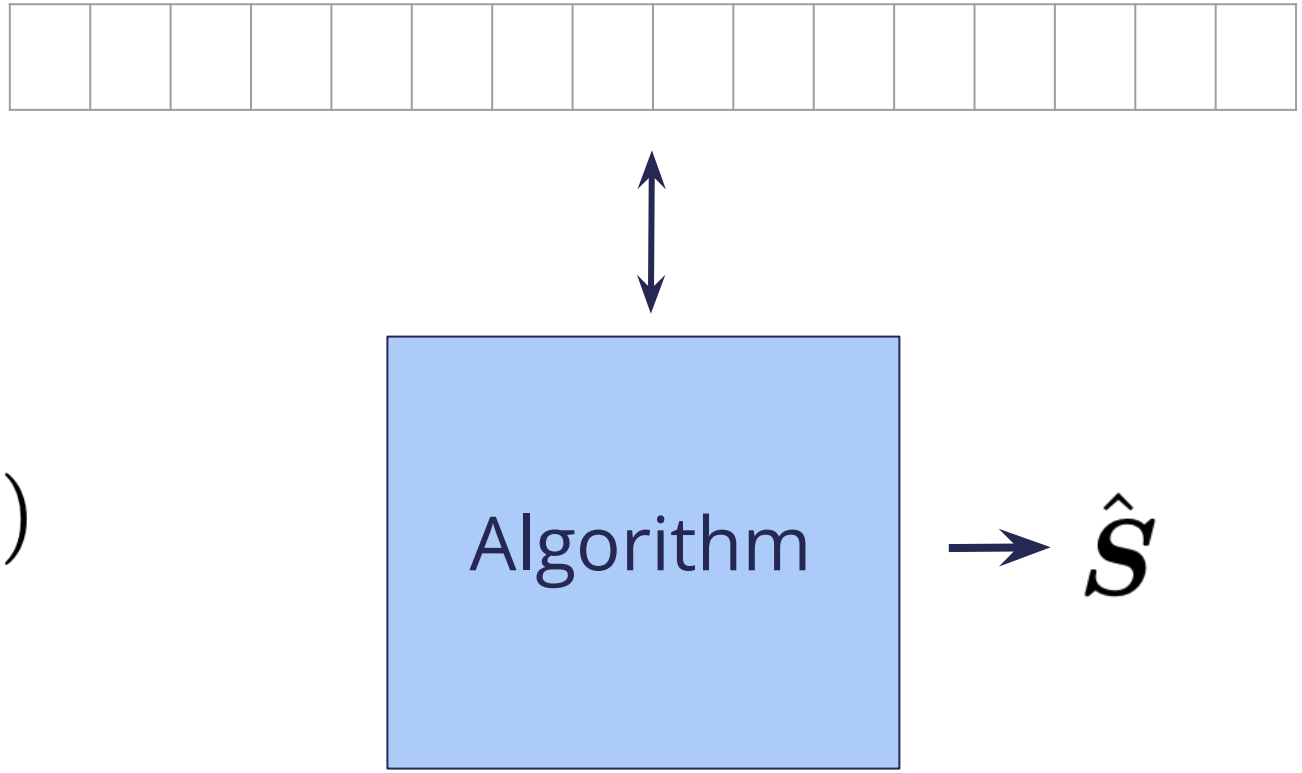
q

Analysis: Union bound over collection.

- Consider single statistic

$$f : \mathcal{X} \to [0, 1]$$

$$\frac{1}{n}\sum_{t=1}^{n} f(x_i) \stackrel{?}{=} \frac{1}{q}\sum_{t=1}^{q} f(x_{i_t})$$



- Failure probability:

$$\Pr\left[|S - \hat{S}| \geq \epsilon\right] \leq 2e^{-2\epsilon^2 q}$$

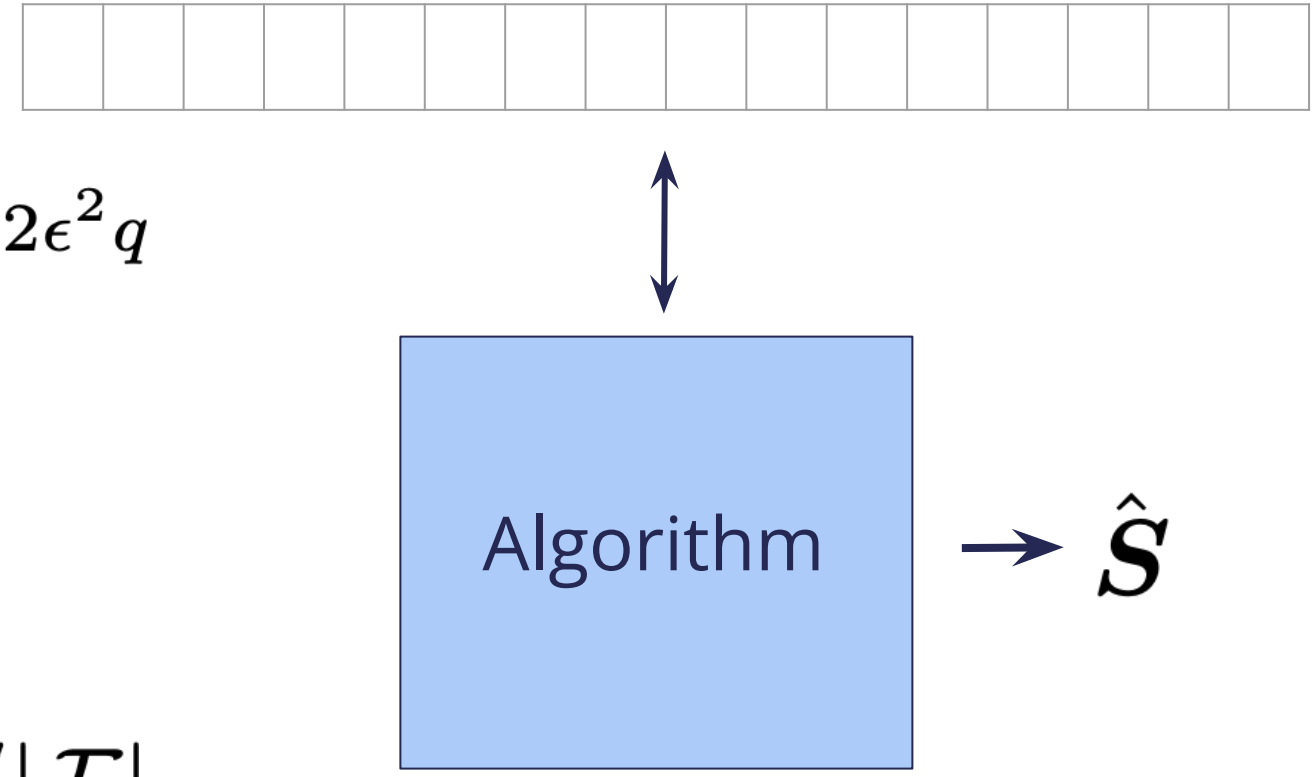
Analysis: Union bound over collection.

- Failure probability:

$$\Pr\left[|S - \hat{\boldsymbol{S}}| \geq \epsilon\right] \leq 2e^{-2\epsilon^2 q}$$

- When $q = \dfrac{\log(2|\mathcal{F}|/\delta)}{2\epsilon^2}$,

$$\leq \delta/|\mathcal{F}|$$

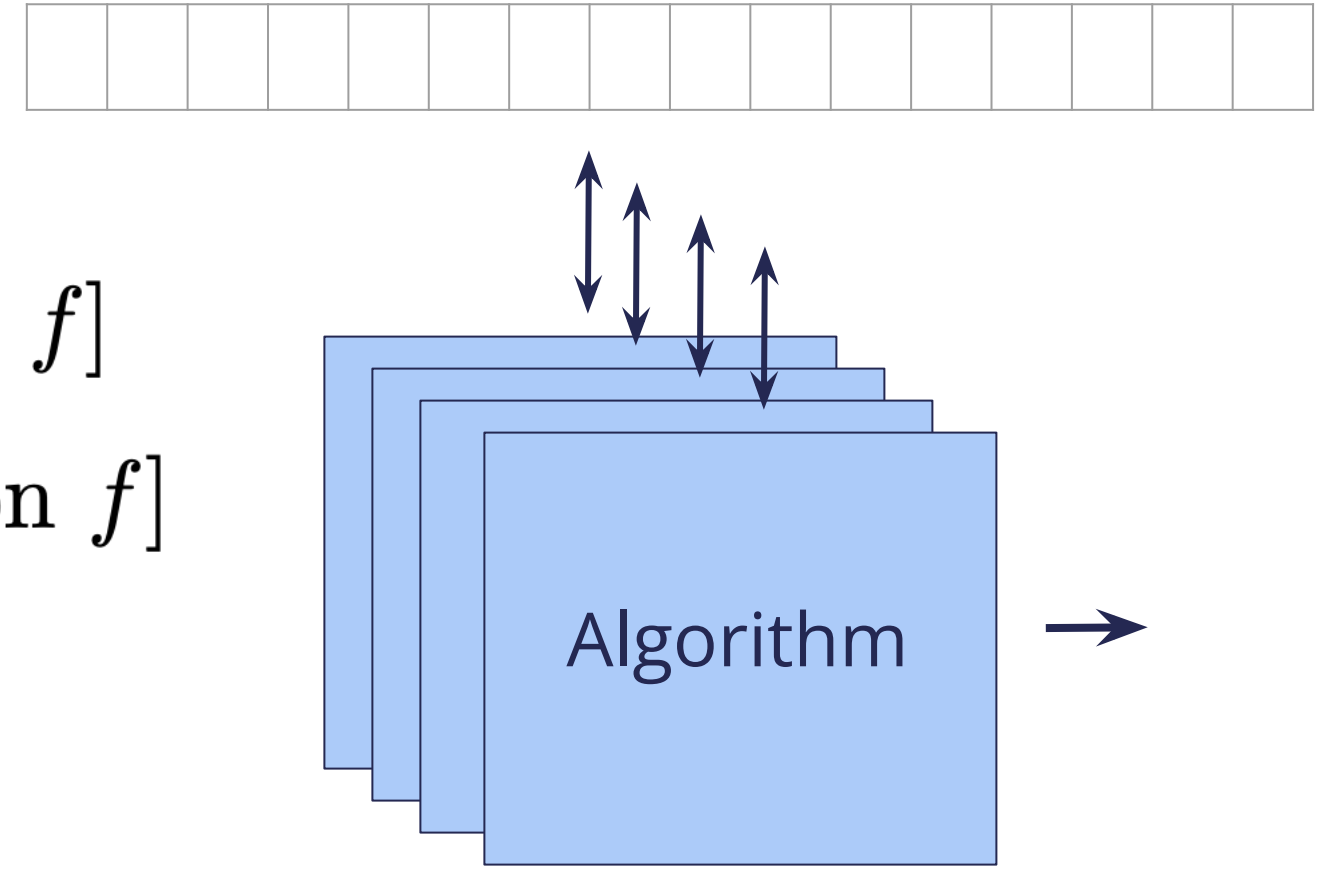


Analysis: Union bound over collection.

- Probability that there exists a mistake:

$$\Pr\left[\exists f \in \mathcal{F} : \text{ mistake on } f\right]$$
$$\leq \sum_{f \in \mathcal{F}} \Pr\left[\text{mistake on } f\right]$$
$$\leq |\mathcal{F}| \cdot \delta / |\mathcal{F}|$$

by union bound



Summary

- Random sampling gives small summaries (“sketches”) of big data
  - For fixed collection of statistics, dramatic data reduction

δ \ ϵ	0.1	0.01	0.001
0.1	413	41K	4.1M
0.01	529	53K	5.3M
0.001	644	64K	6.4M

- Important caveat: functions must be fixed in advanced!